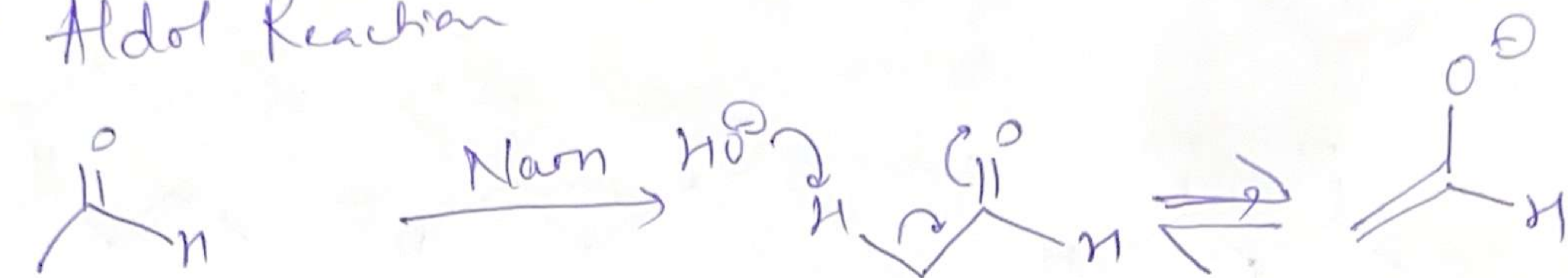
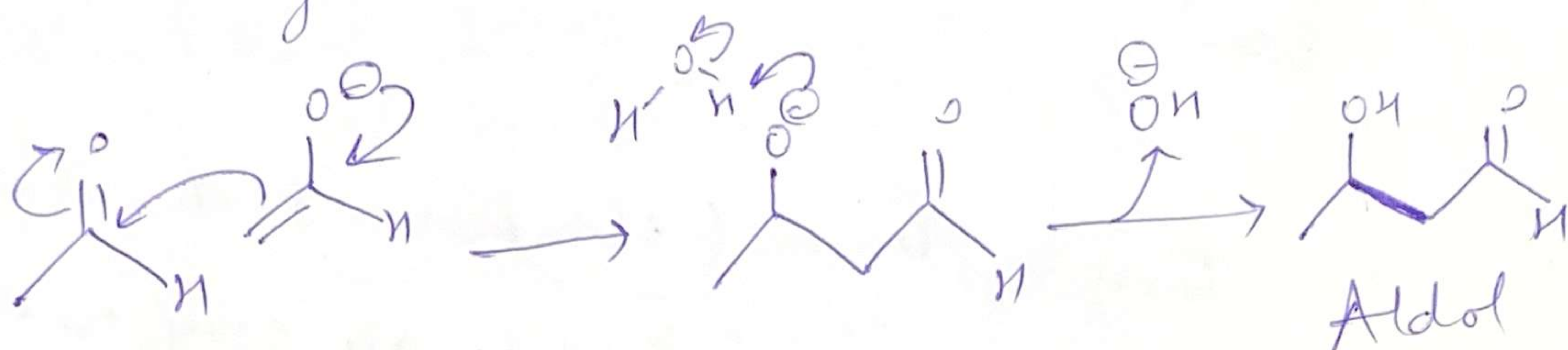


①

① Aldol Reaction



Acetaldehyde



Aldol

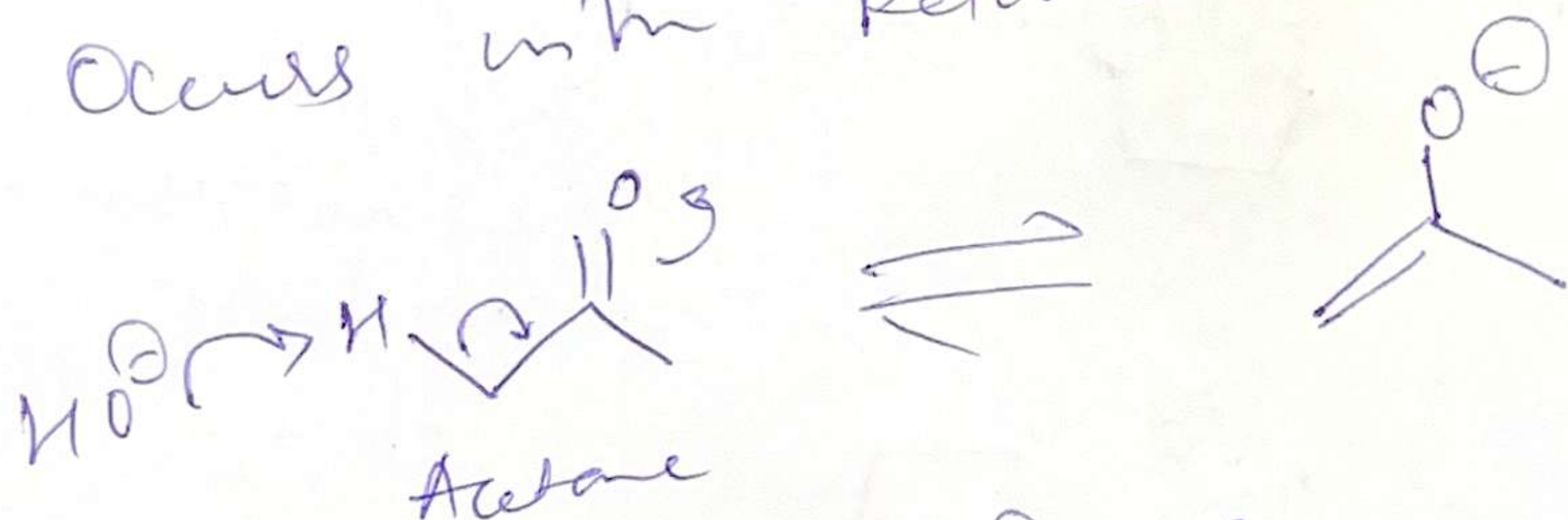
3-hydroxy
Butanal ~~butanal~~

Base is regenerated in last step.

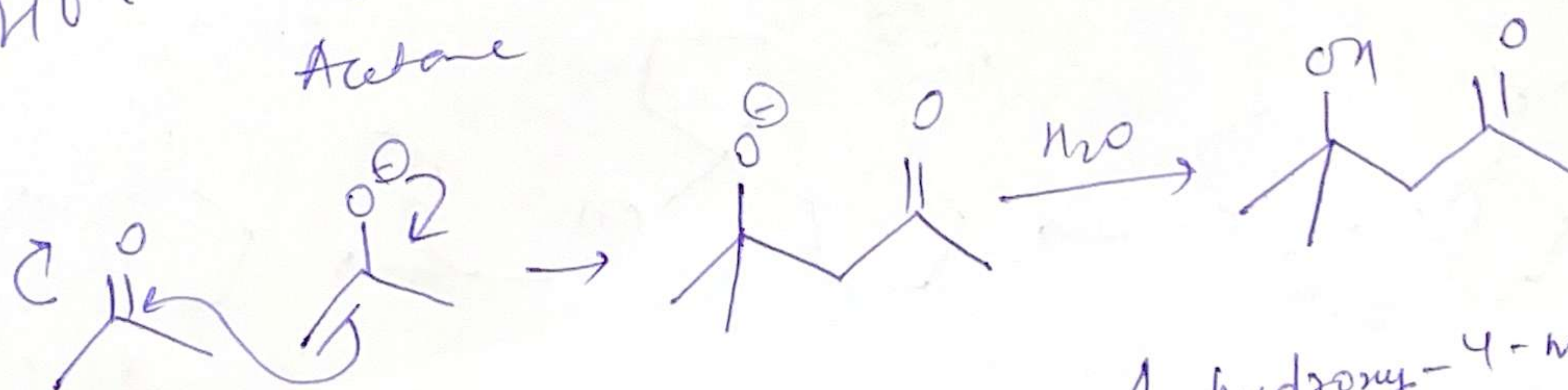
②

Therefore, it's a catalyst.

② Occurs with ketones as well.

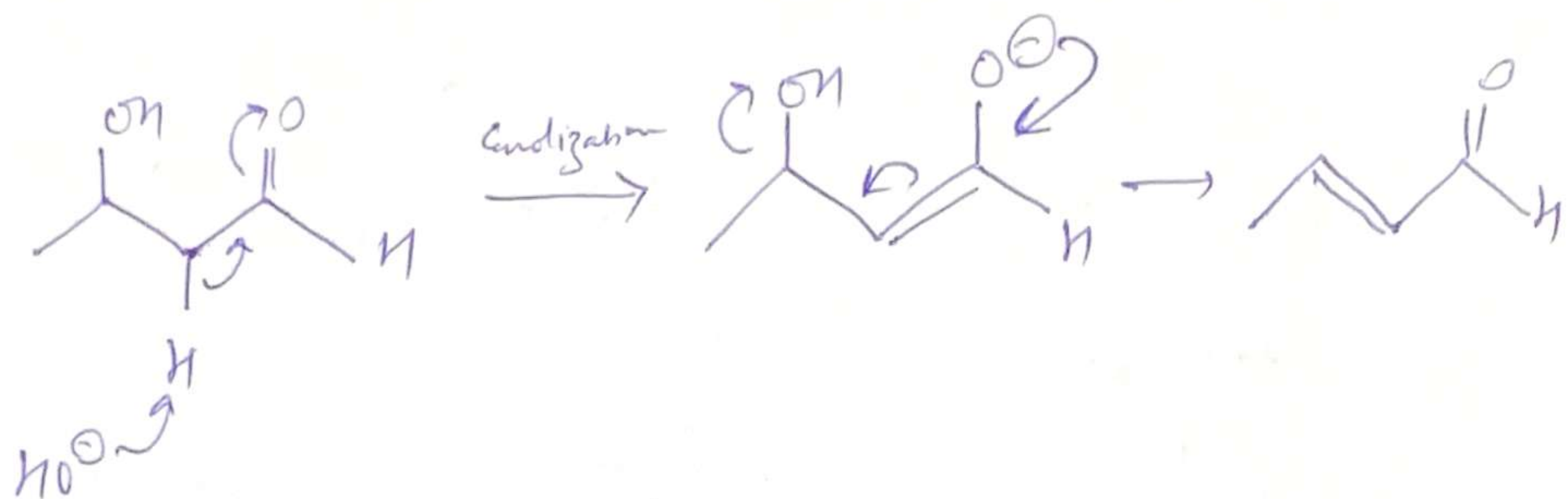


Acetone



4-hydroxy-4-methyl
pentan-2-one

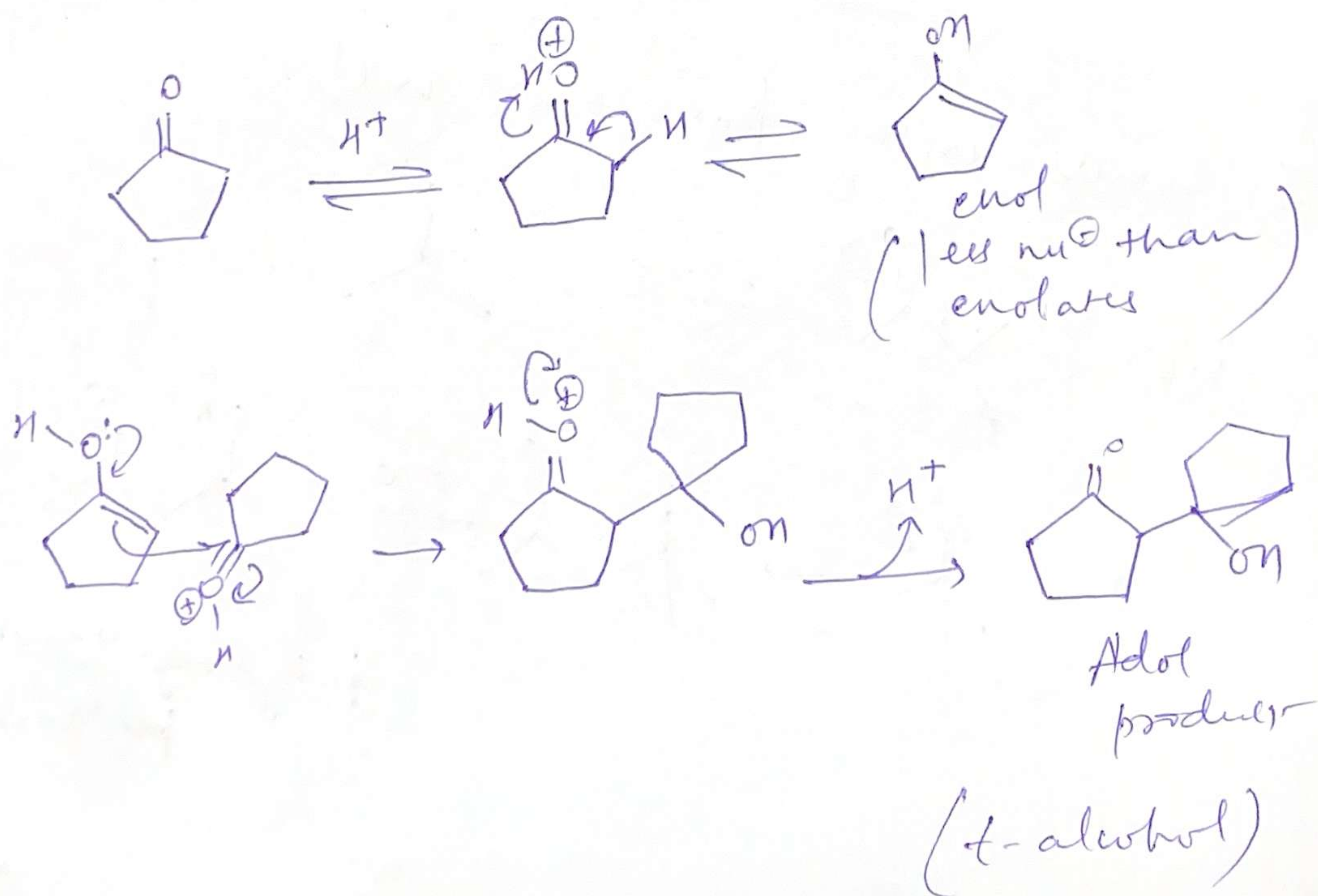
If base is present in excess :-



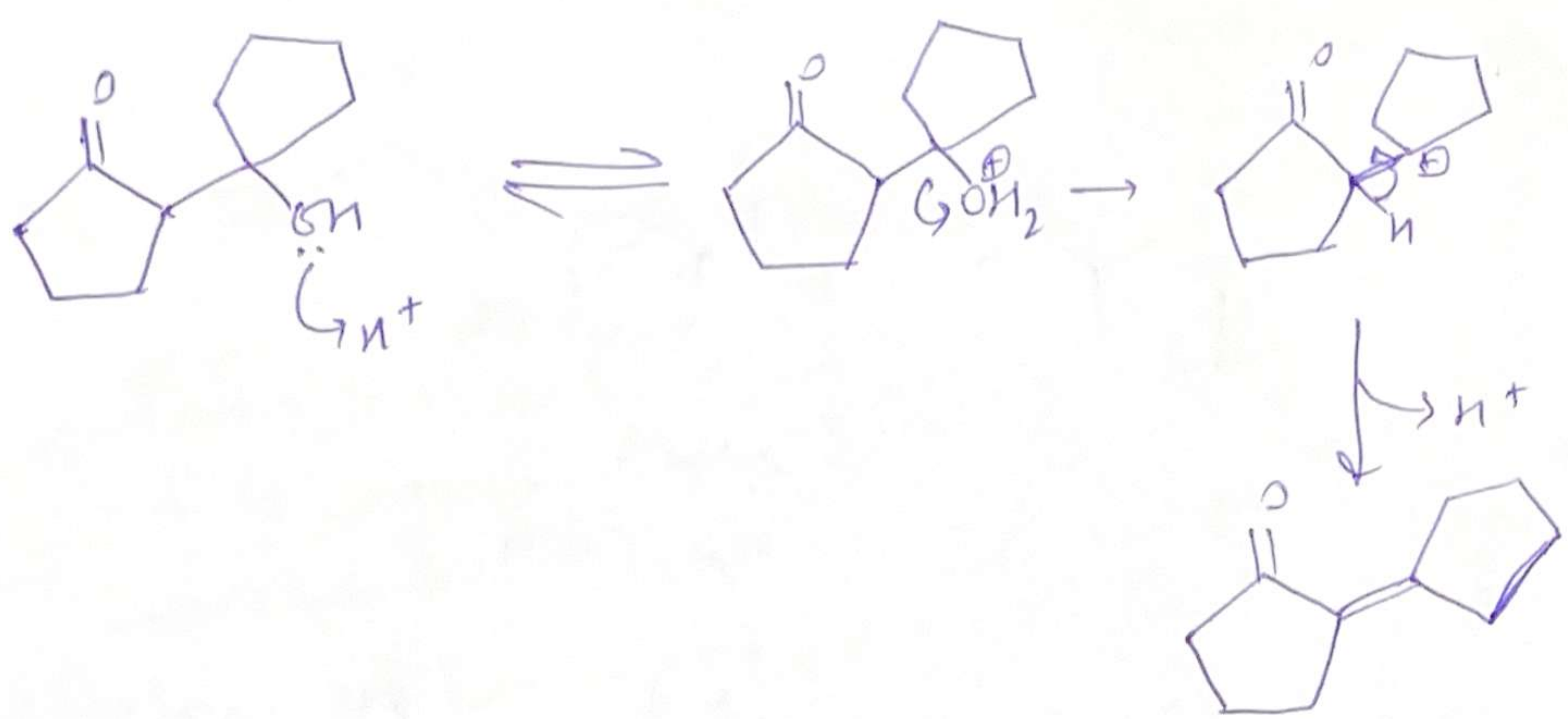
Base → Excess (elimination product)

Structure → Some compd. are easy to stop at the aldol stage

Acid-catalyzed aldol reactⁿ commonly give unsaturated products instead of aldols

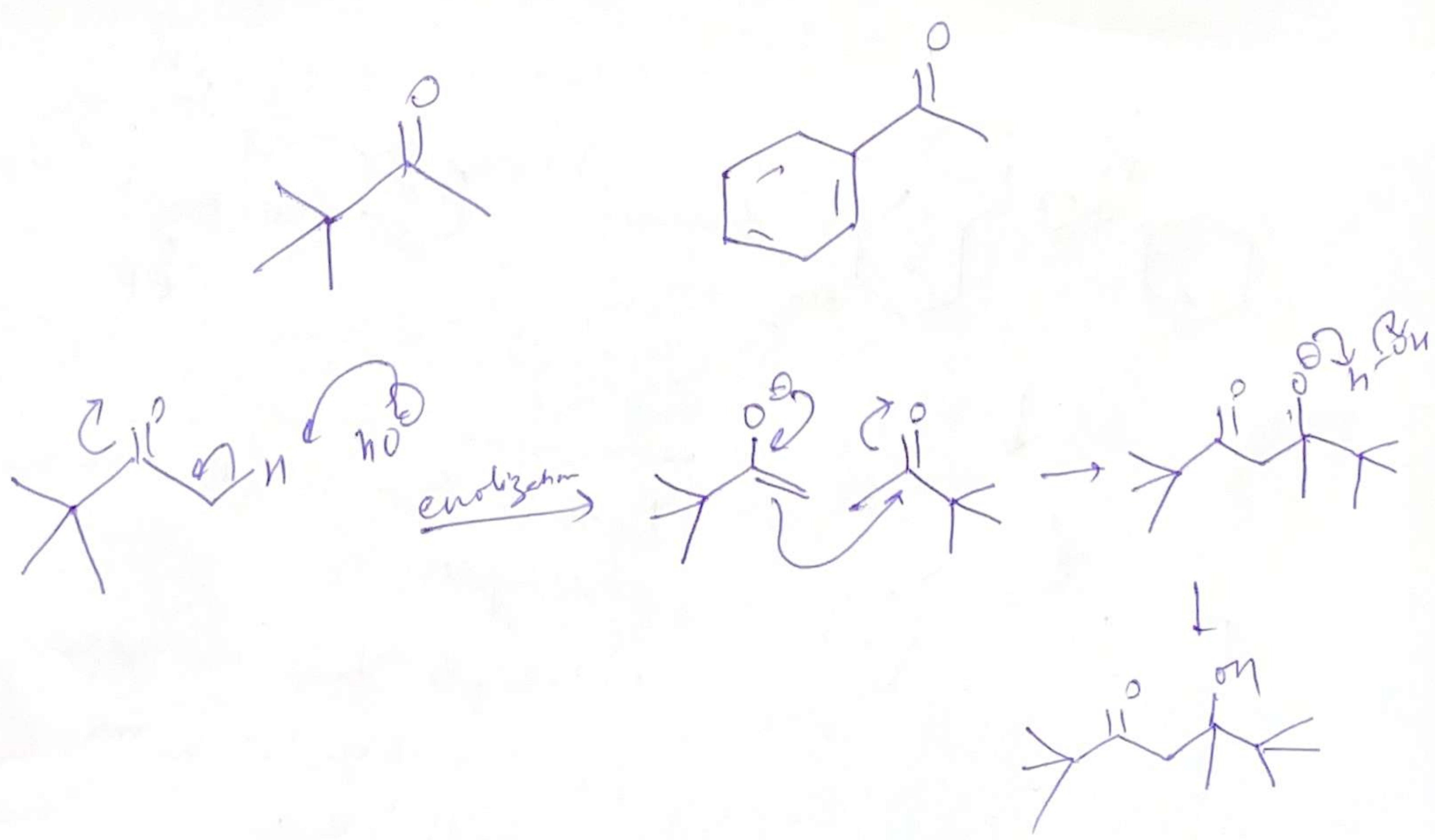


②

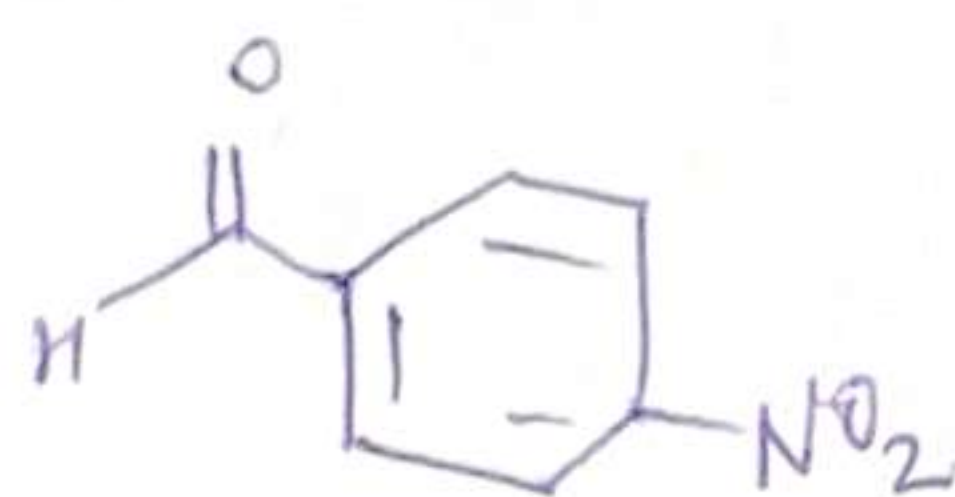
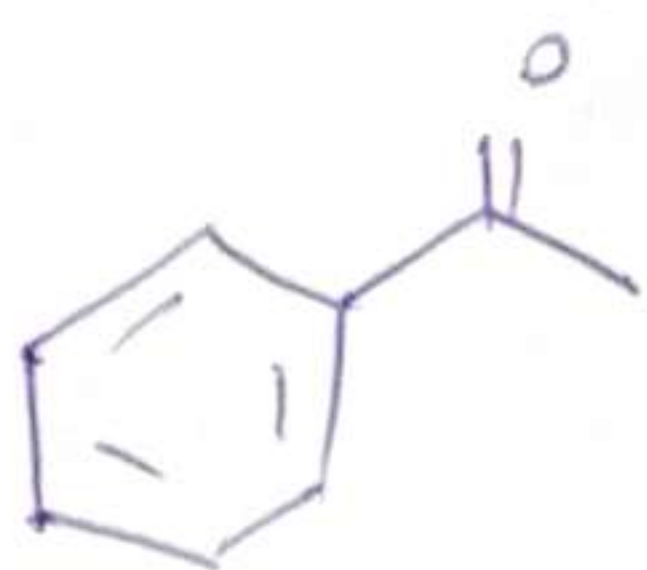


Aldol Condensation → Loss of small molecule usually water

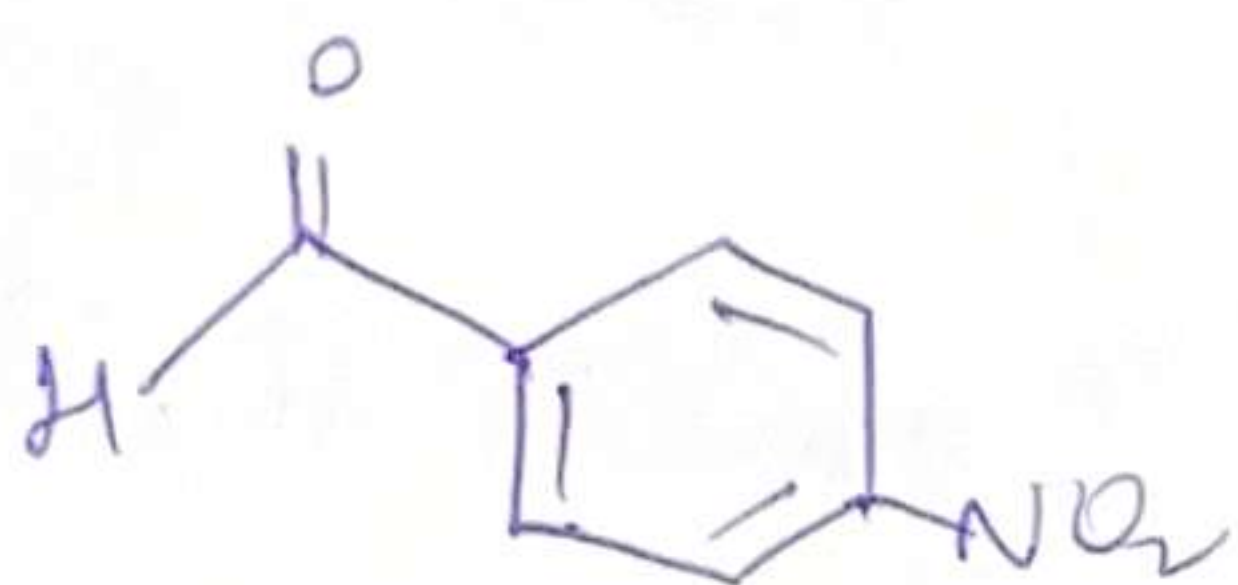
Aldol react of unsymmetrical ketones



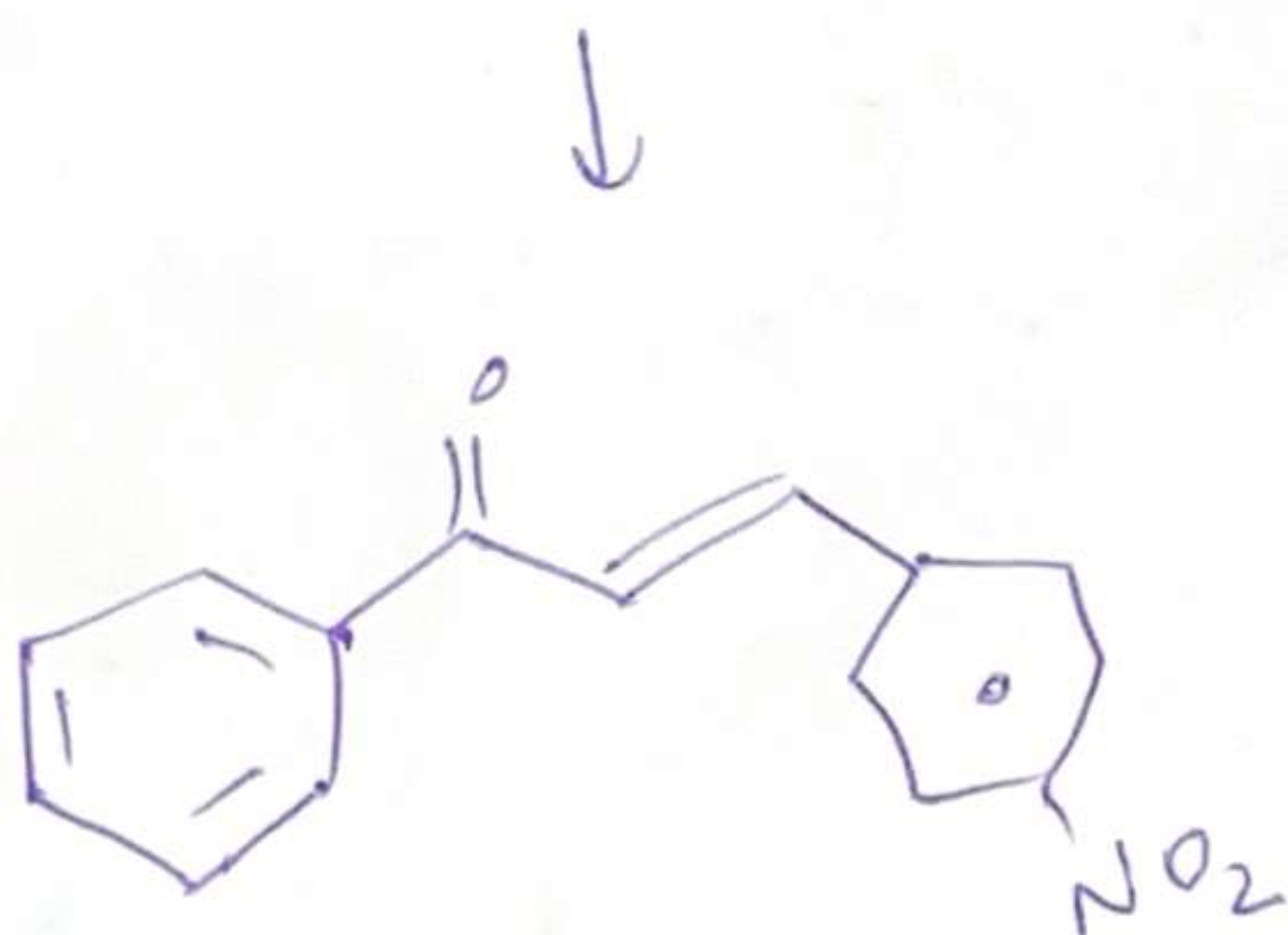
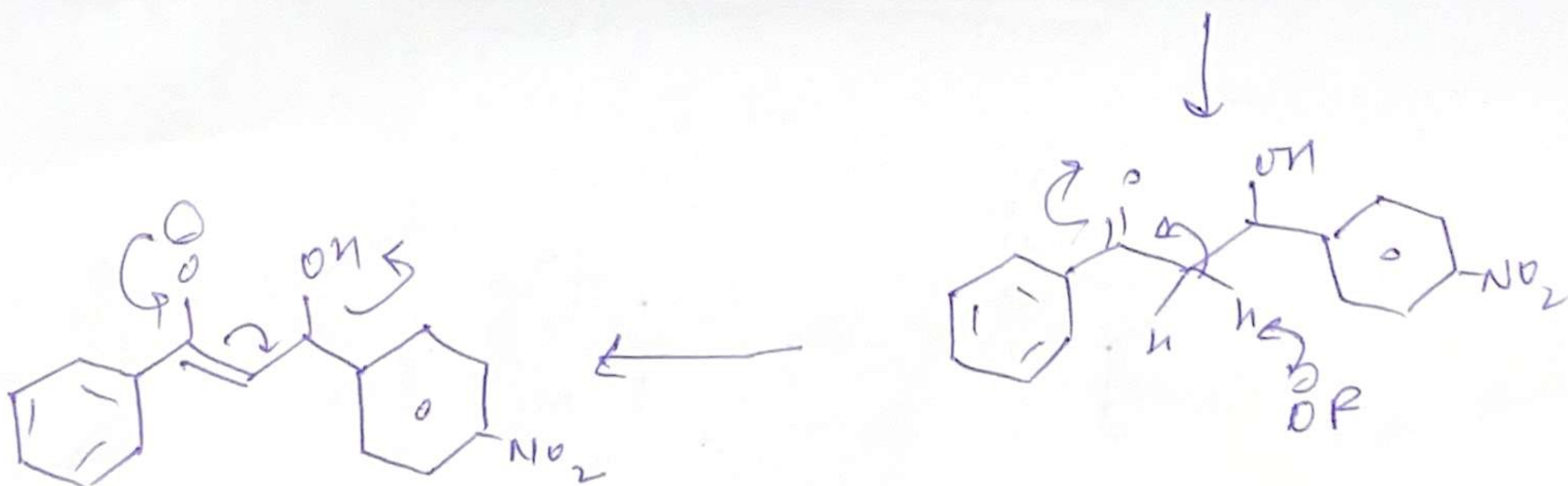
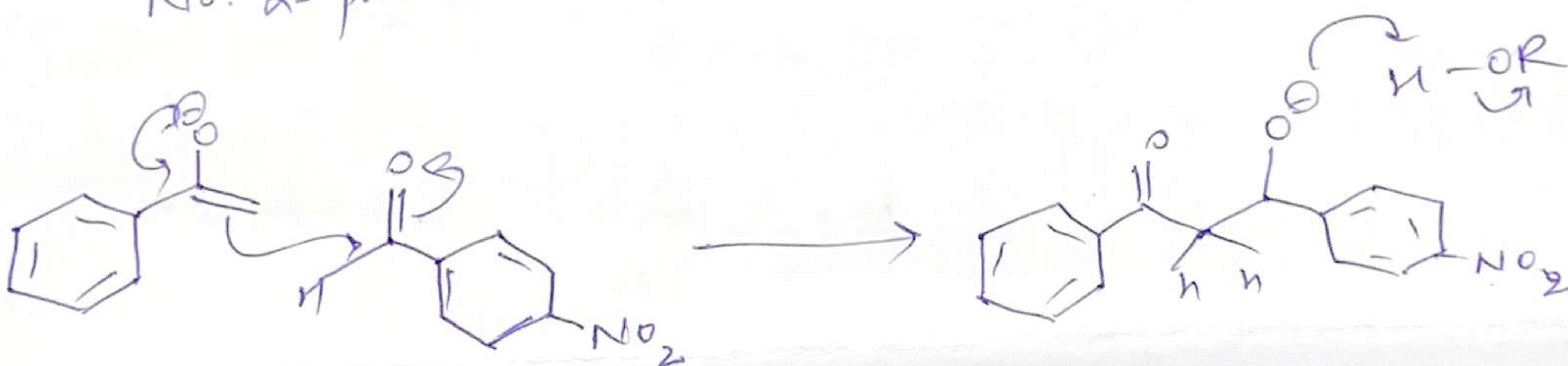
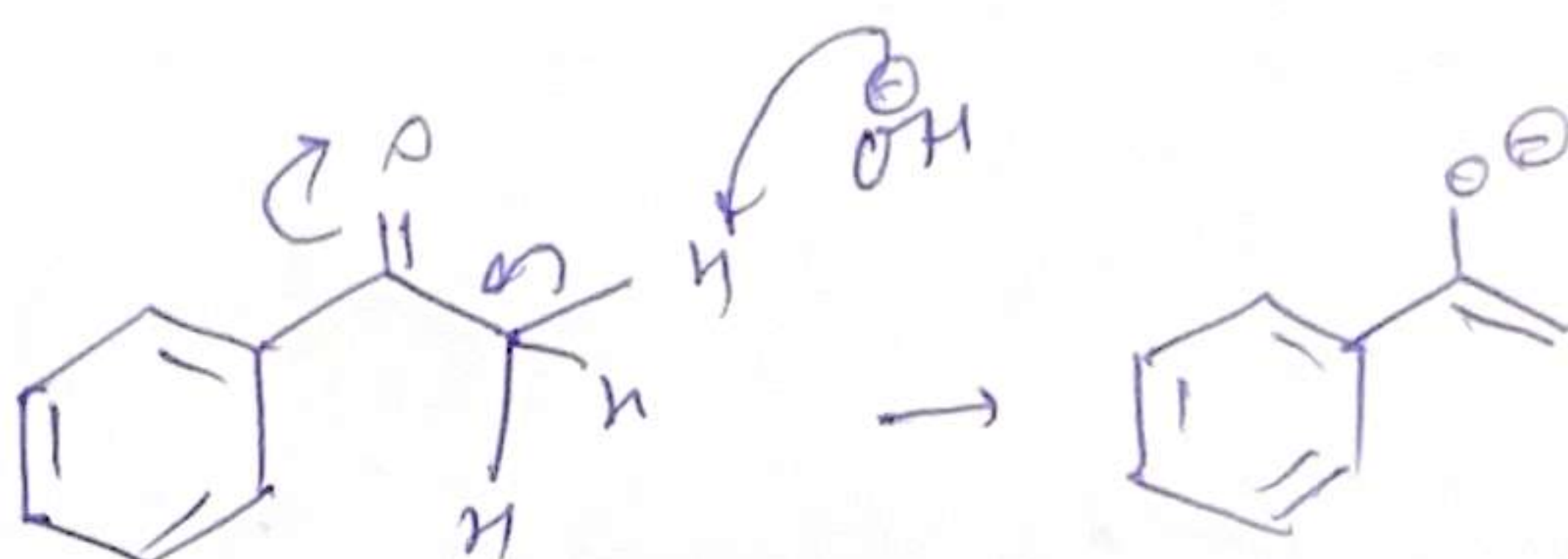
Cross-Condensation:-



↓ NaOH
H₂O/EtOH



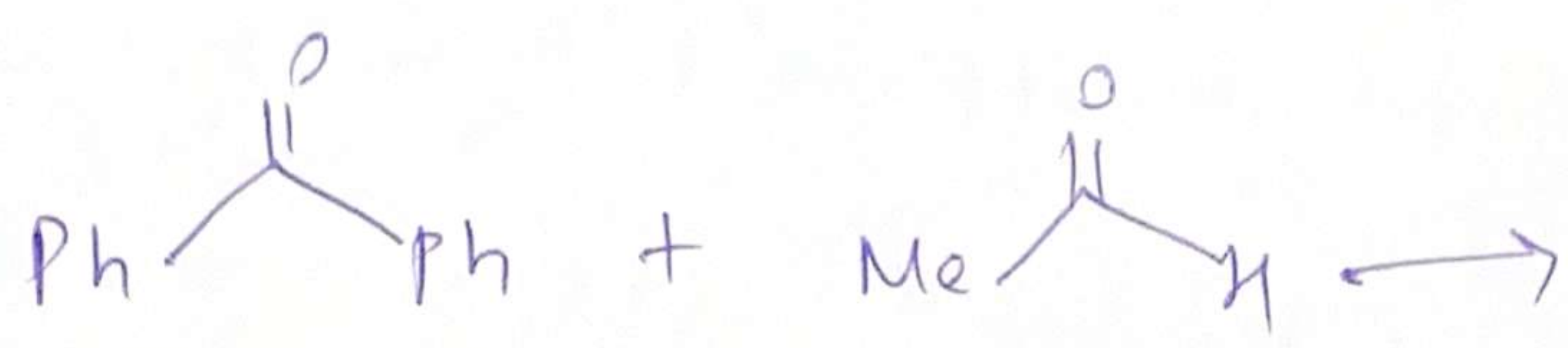
No. α -protons



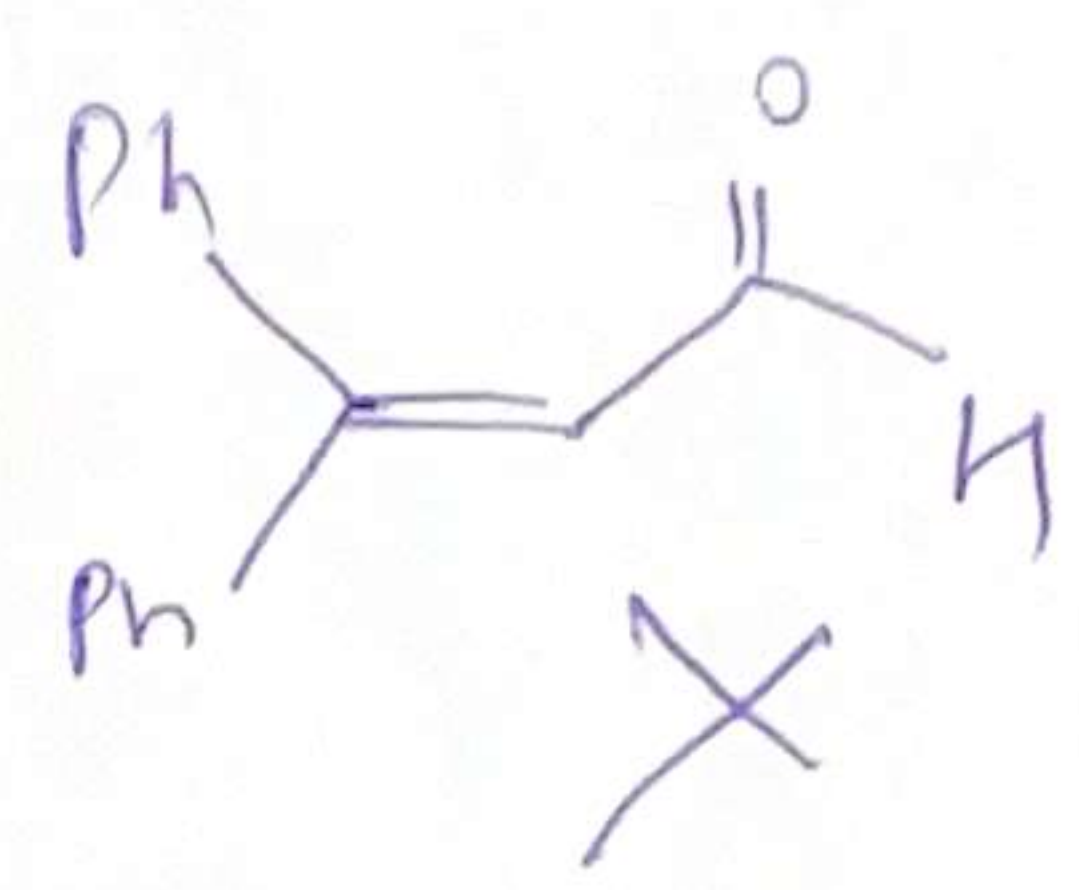
Ketones less reactive than aldehydes

Aldehyde has an ~~electro~~ EWG making it more electrophilic.

Enolate selects the better electrophile i.e. aldehyde



- Ketone is very hindered & conjugated.
- less electrophilic
- Therefore enolate will choose the best electrophile & attack aldehyde.

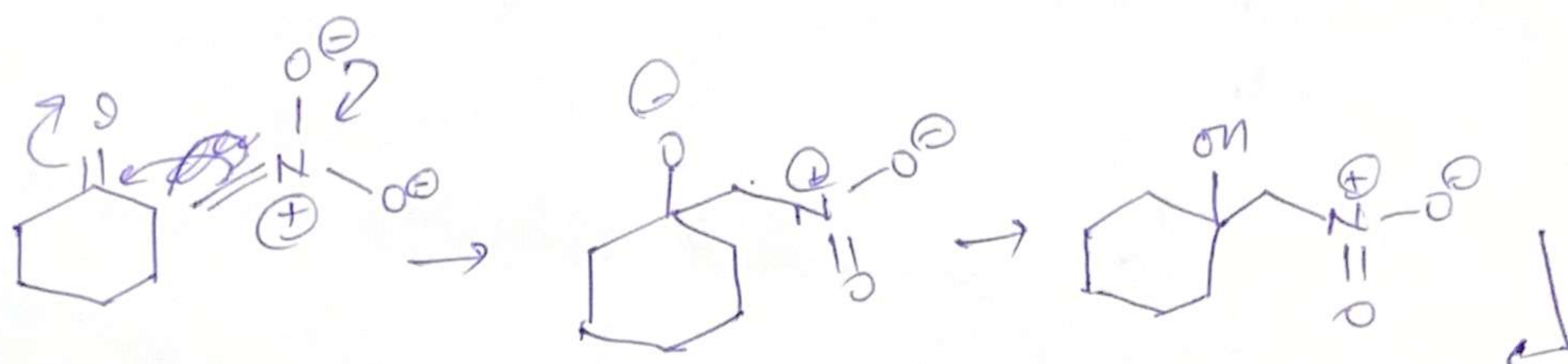
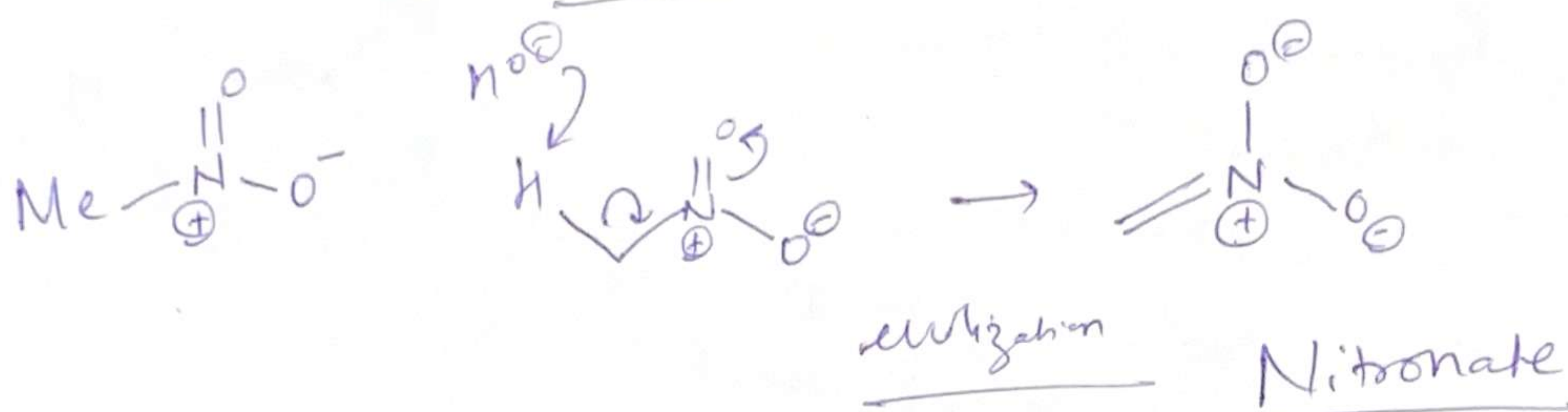


For Successful crossed aldol reaction

- ① only one partner must be capable of enolization.
- ② The other partner must be incapable of enolization & be more electrophilic than the enolizable partner.

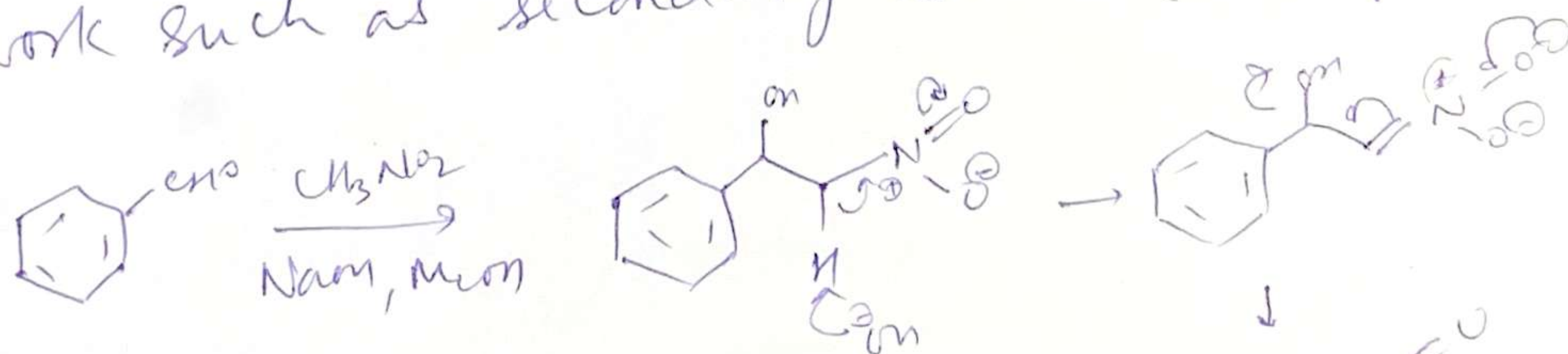
Compounds that can enolize but are not electrophilic

Henry's Reaction

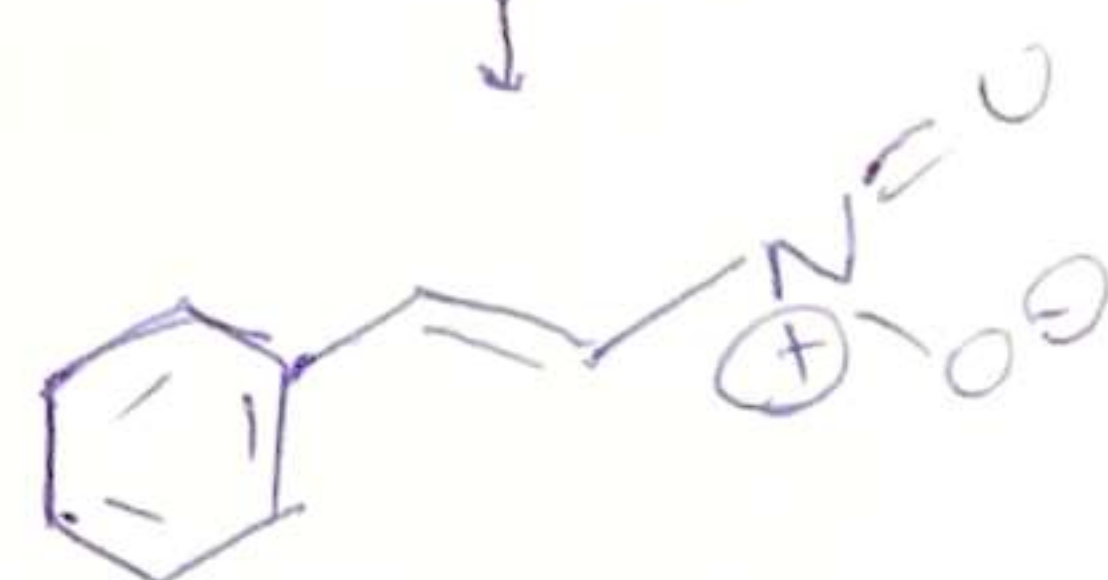


self aldol of ketone doesn't happen because
 base prefers to remove proton from nitromethane
 pK_a of ketone = 20
 Nitromethane = 10
 NaOH ($\text{pK}_a = 15.7$)

Not necessary to use NaOH . Even amines can
 work such as secondary amines (mostly used)



Difficult to avoid Elimination
 step specially with Aryl Carbonyls.



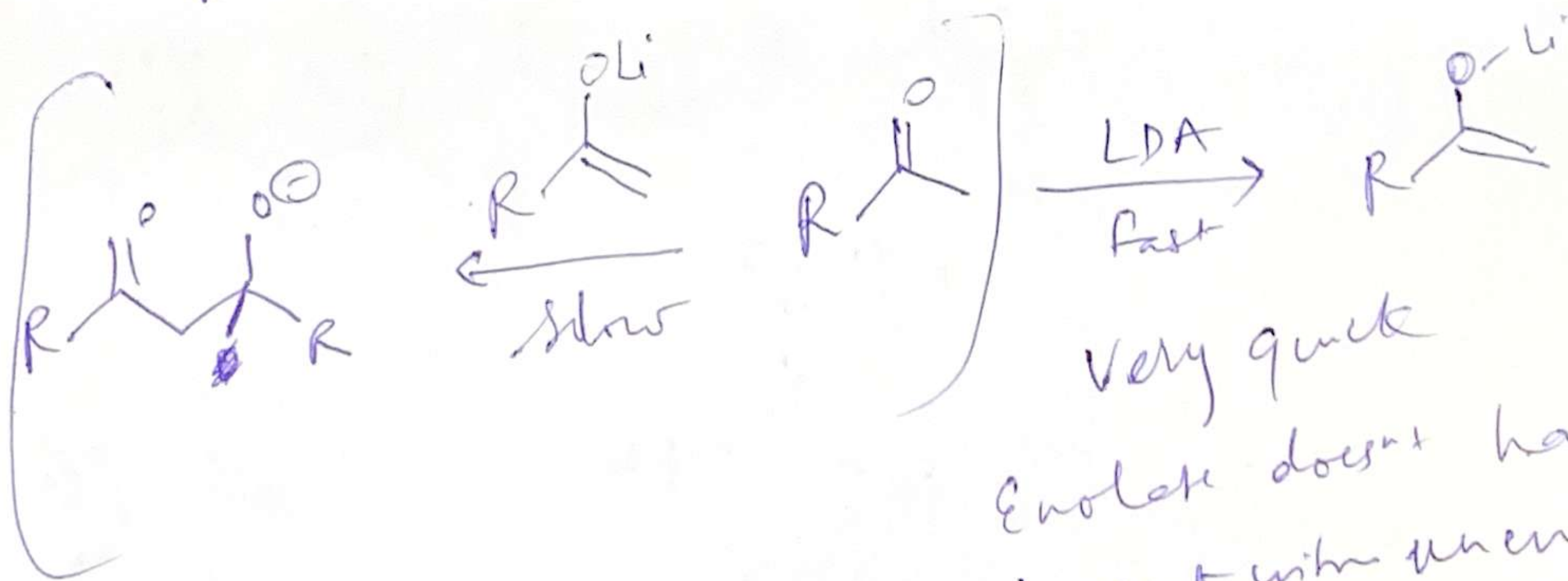
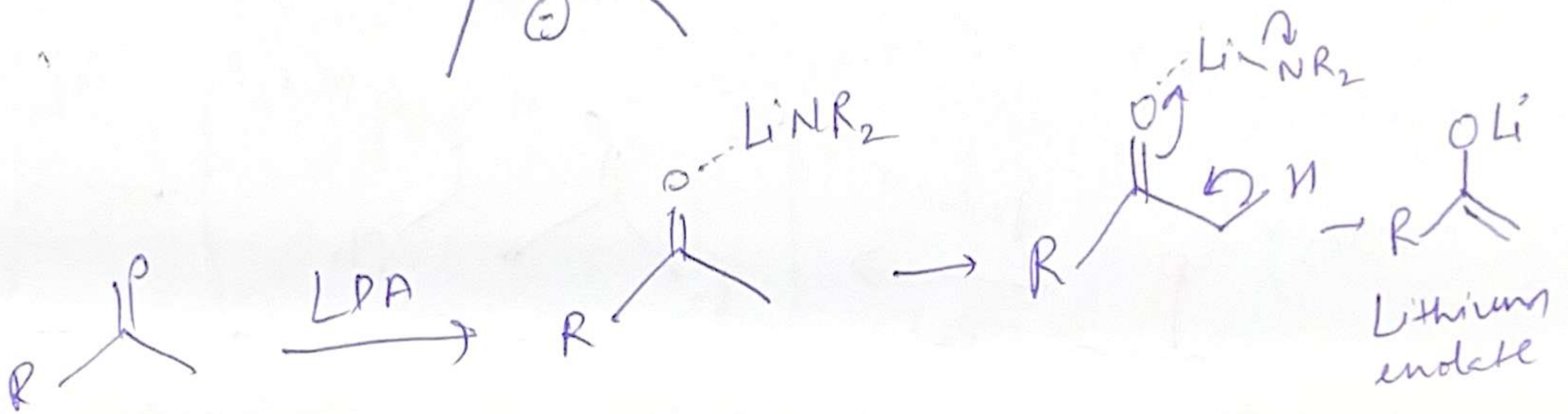
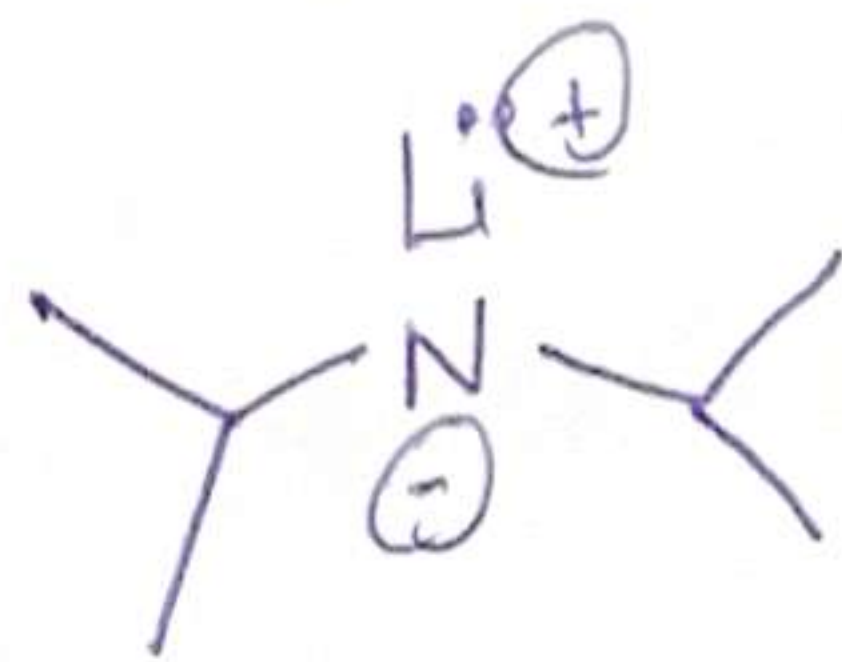
Cross aldol

(4)

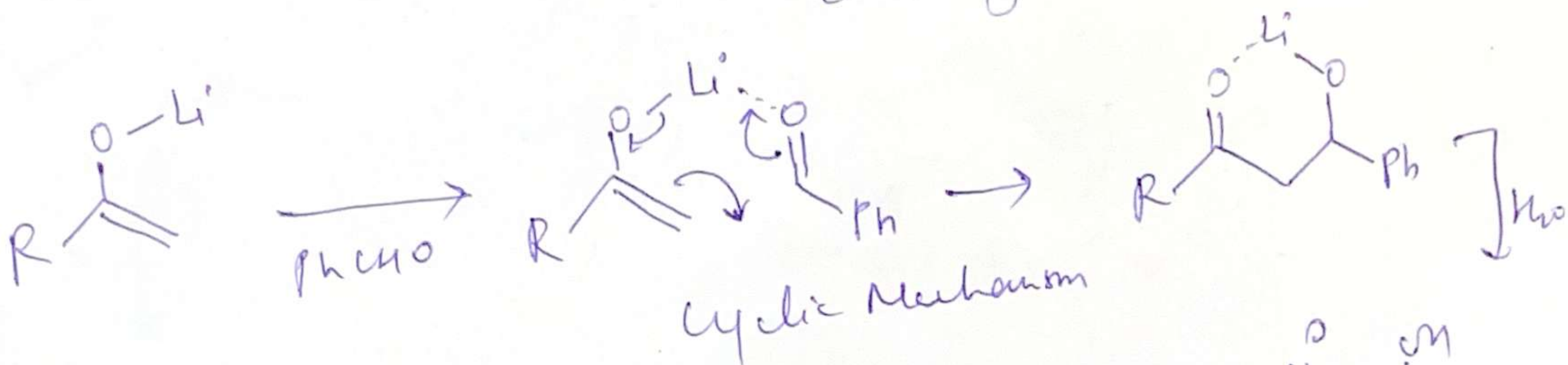
- only one enolizable component
- only one set of enolizable protons
- A carbonyl electrophile more reactive than the compound being enolized.

In case which fail to meet these criteria, specific enol equivalents must be used.

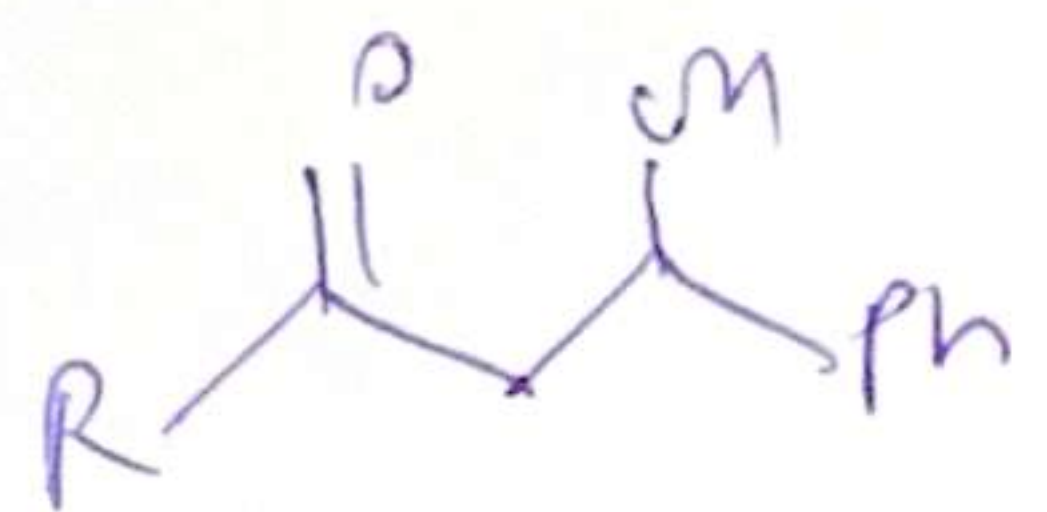
① Lithium enolates



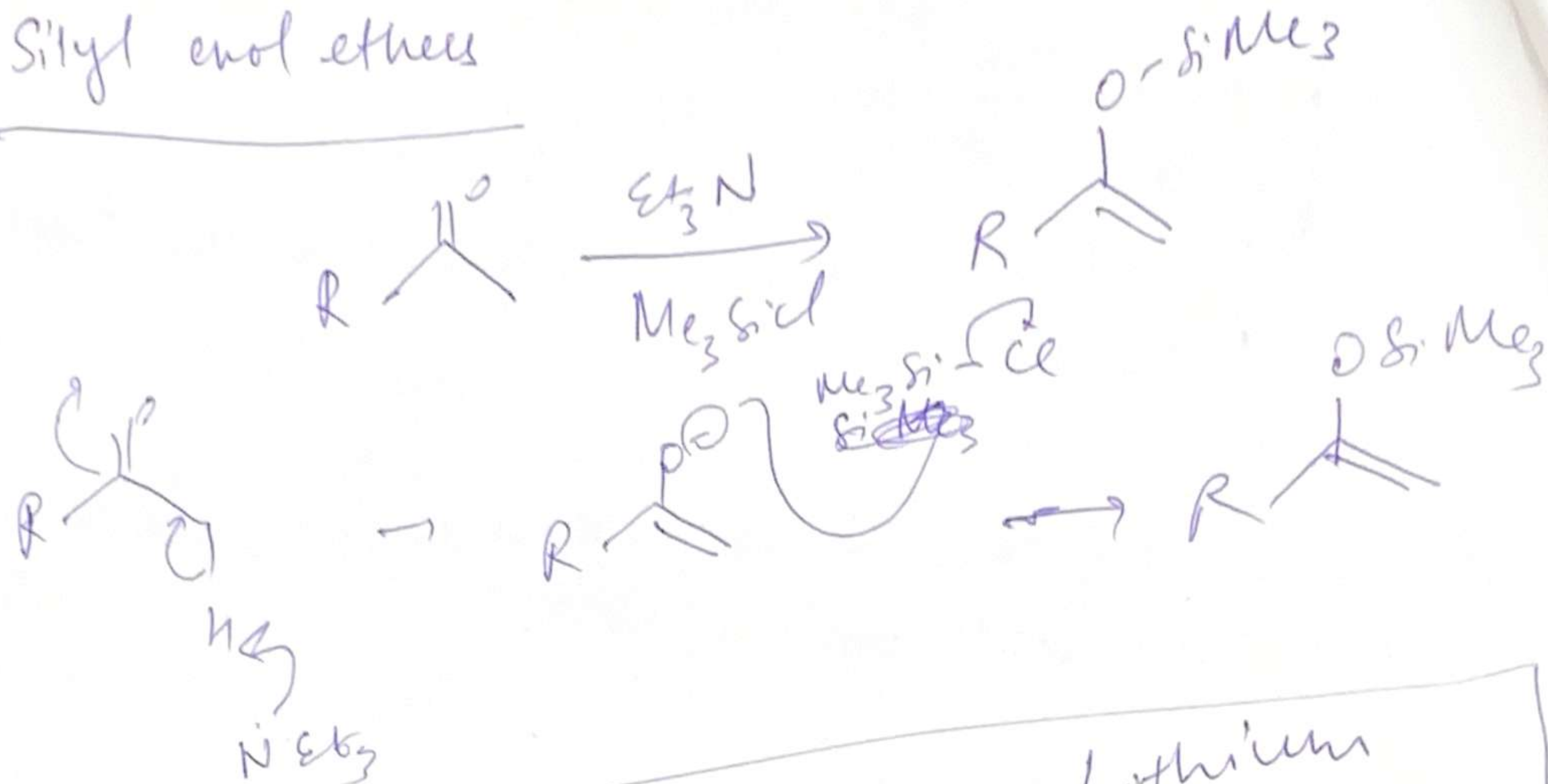
Very quick
Enolate doesn't have chance
to react with unenolized
carbonyl compound.



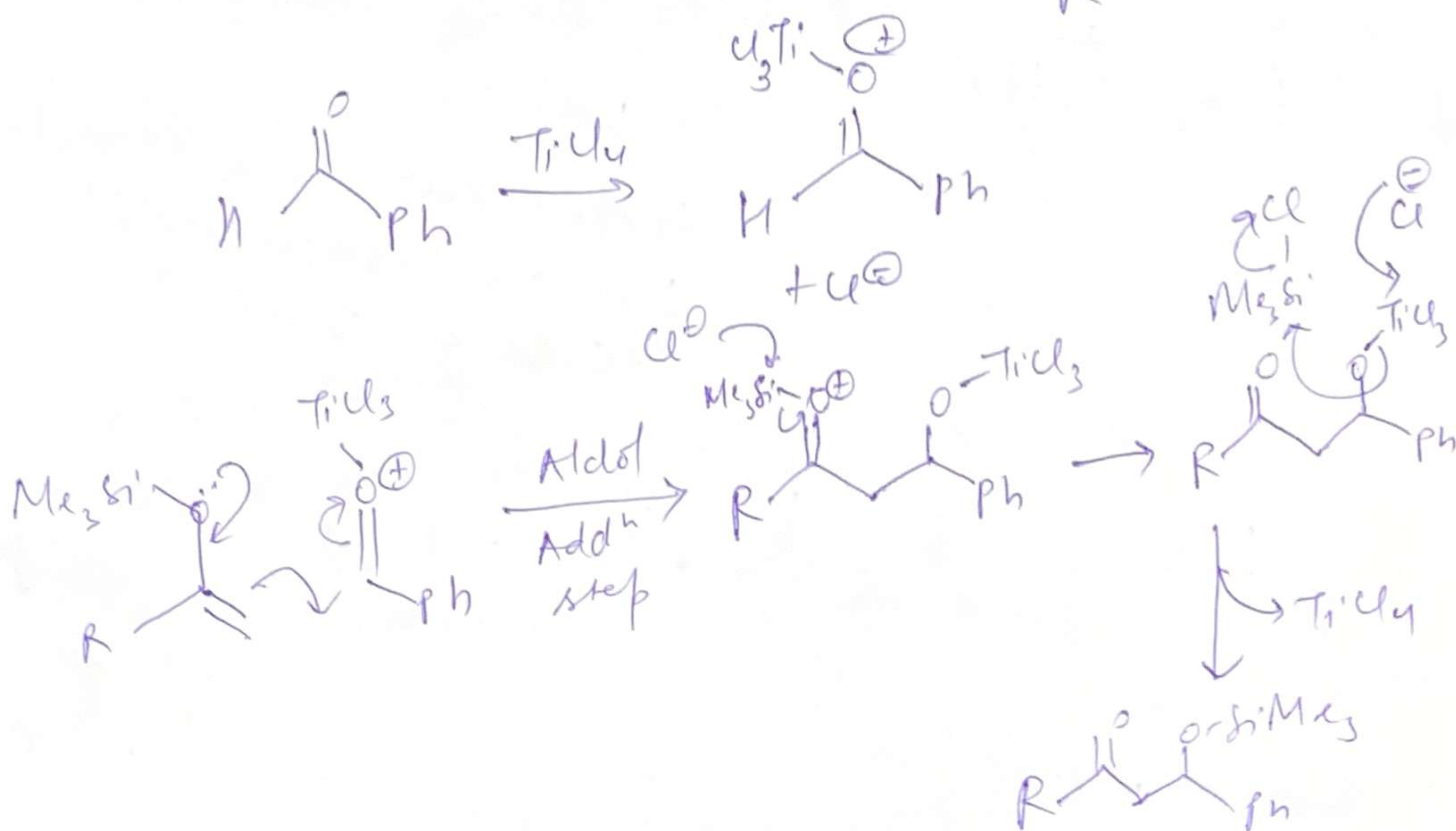
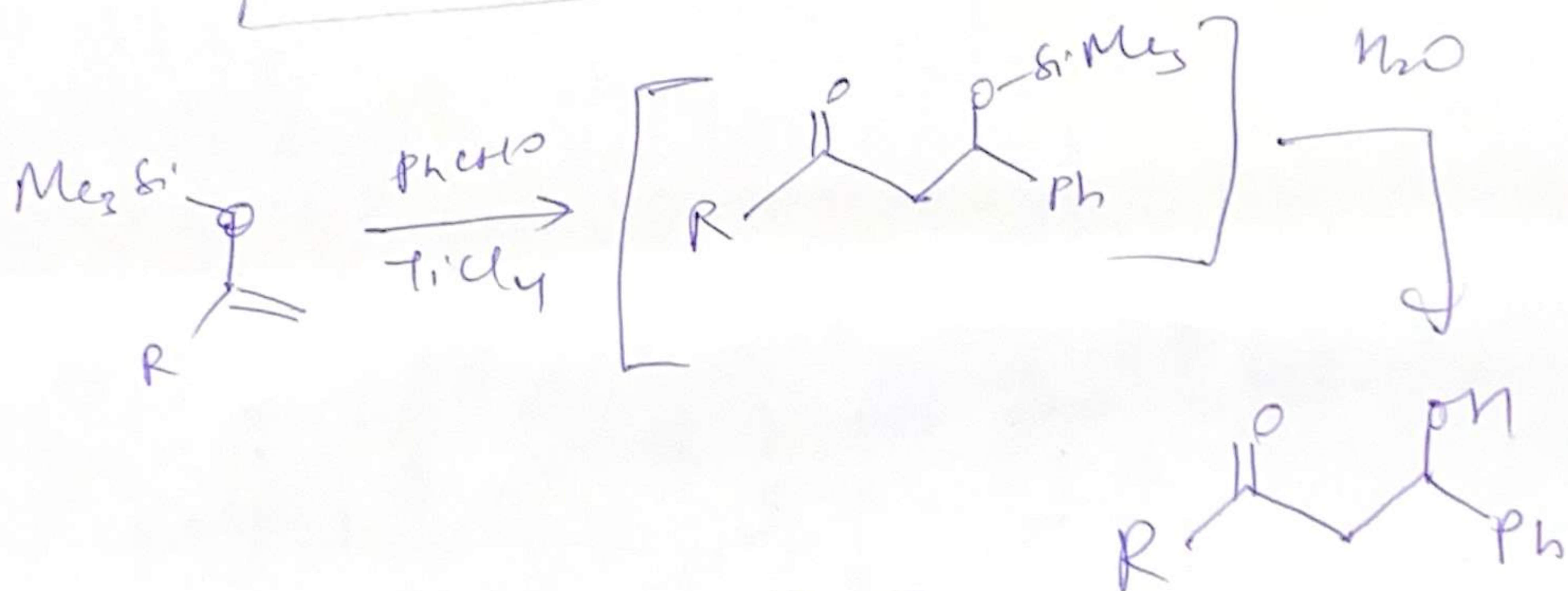
Works well even if the
electrophilic partner is enolizable



Silyl enol ethers

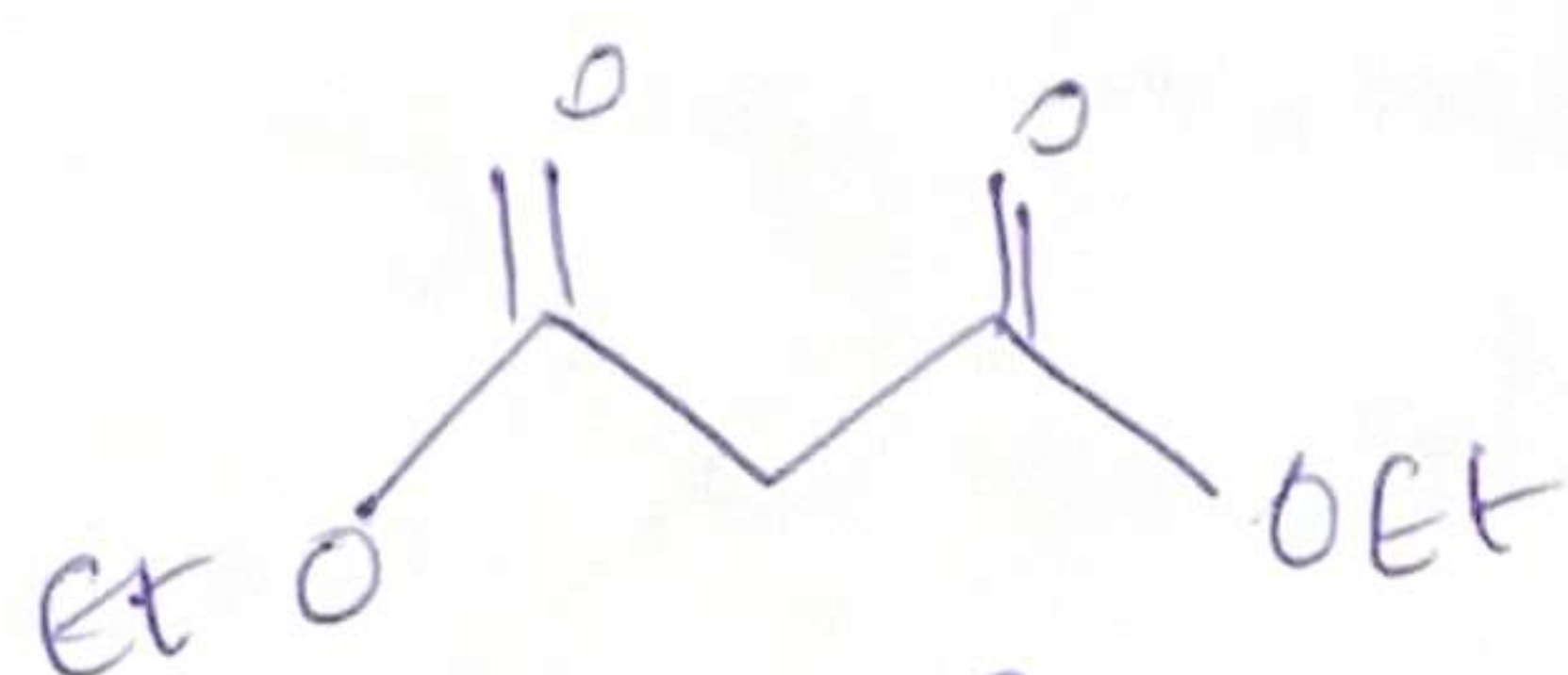


Less reactive than Lithium enolates

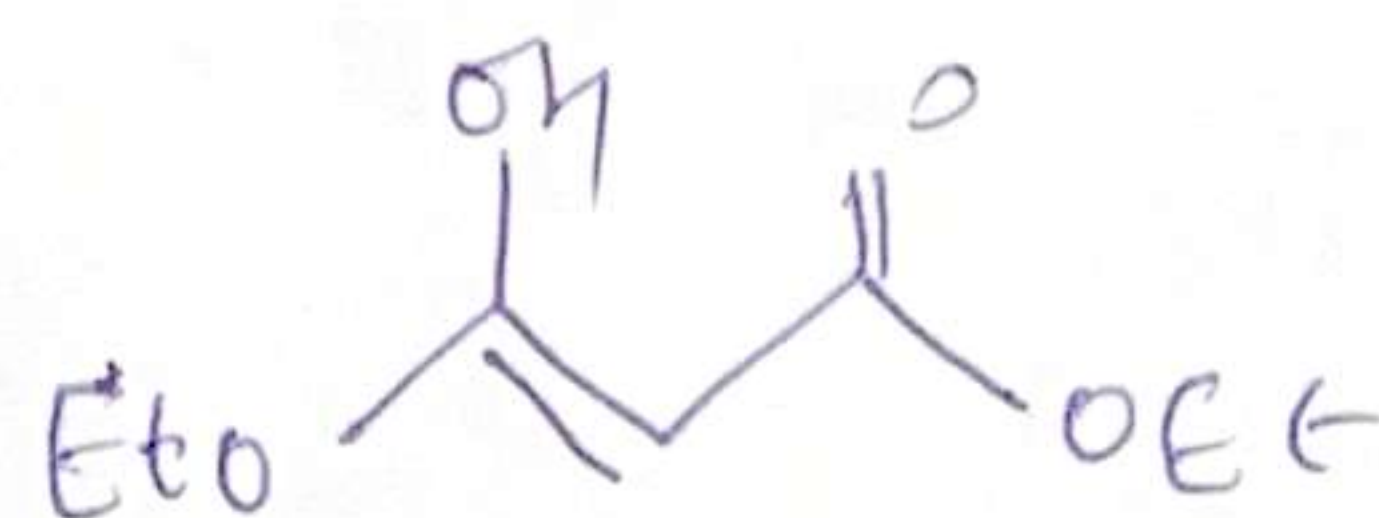


1,3-di Carbonyl ~~Carbonyl~~ Compounds

(5)

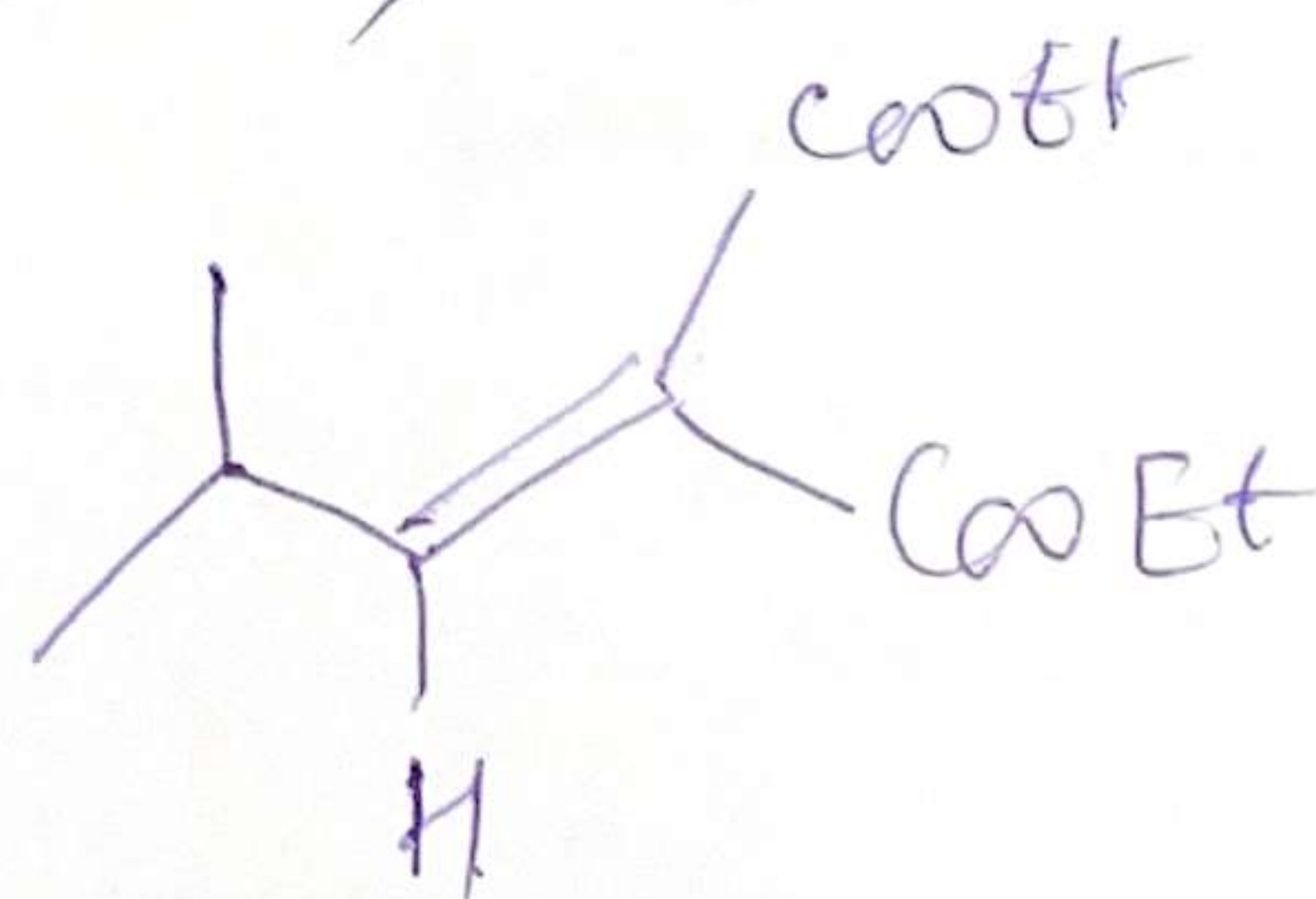
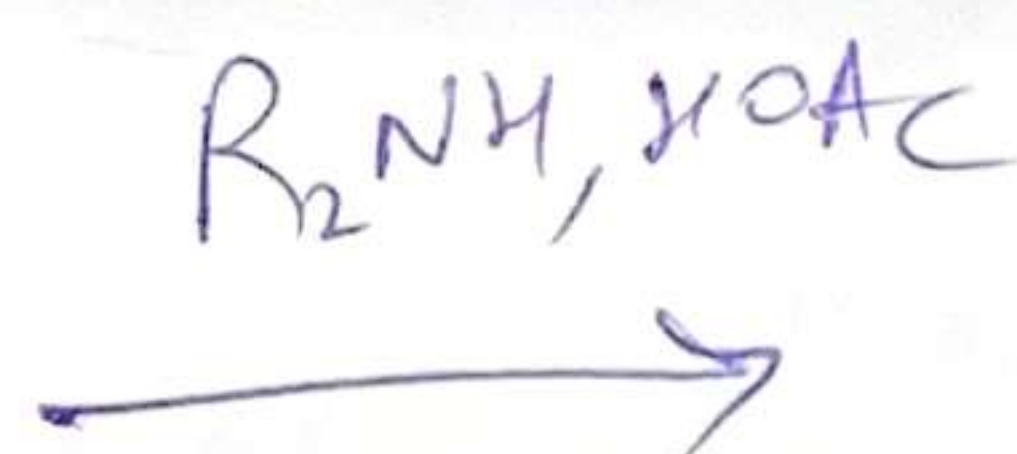
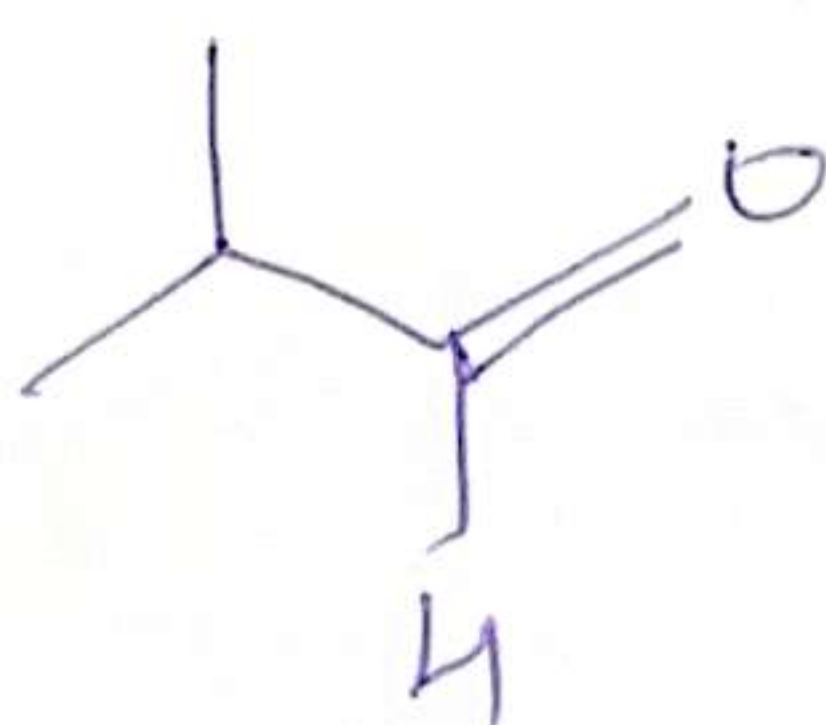
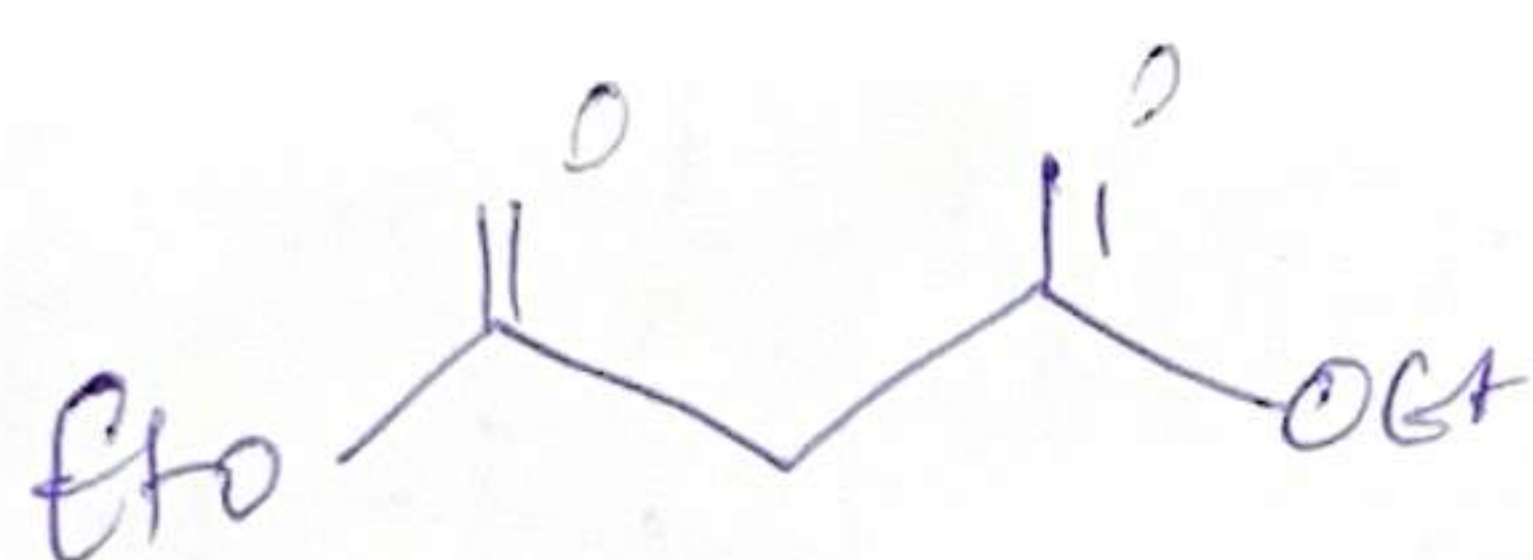


Diethyl
malonate



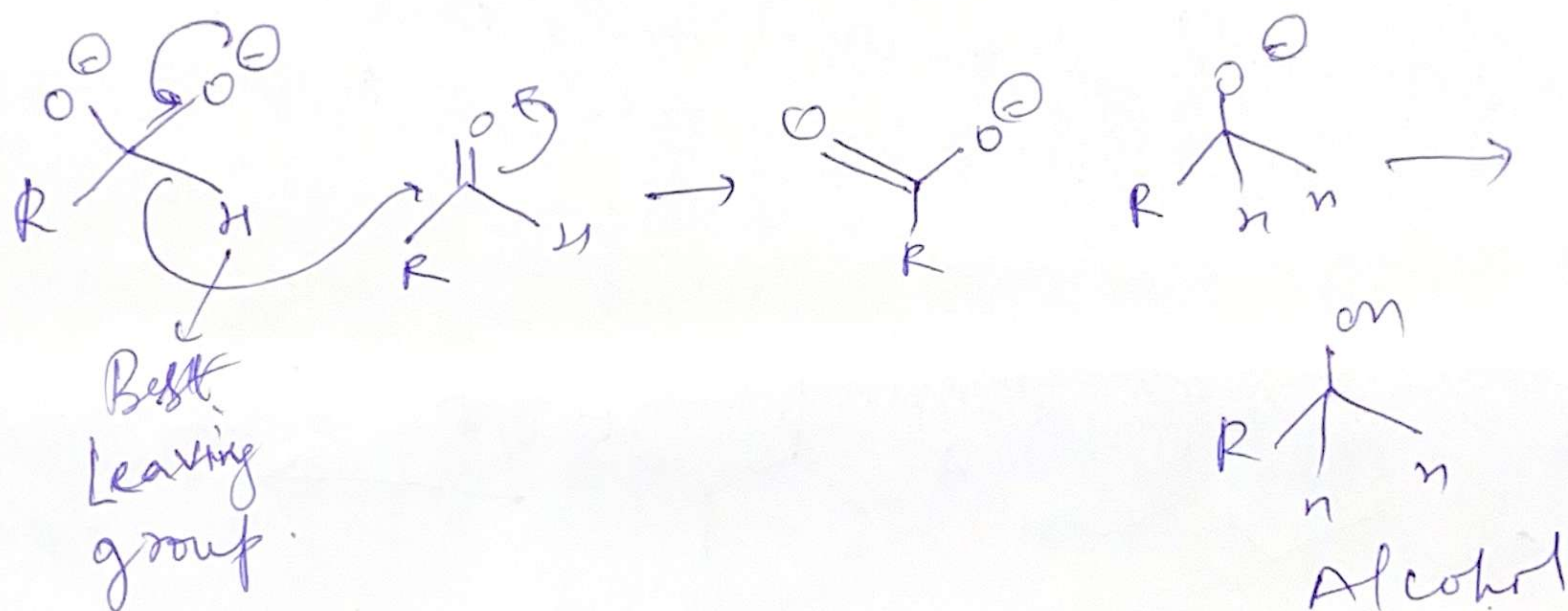
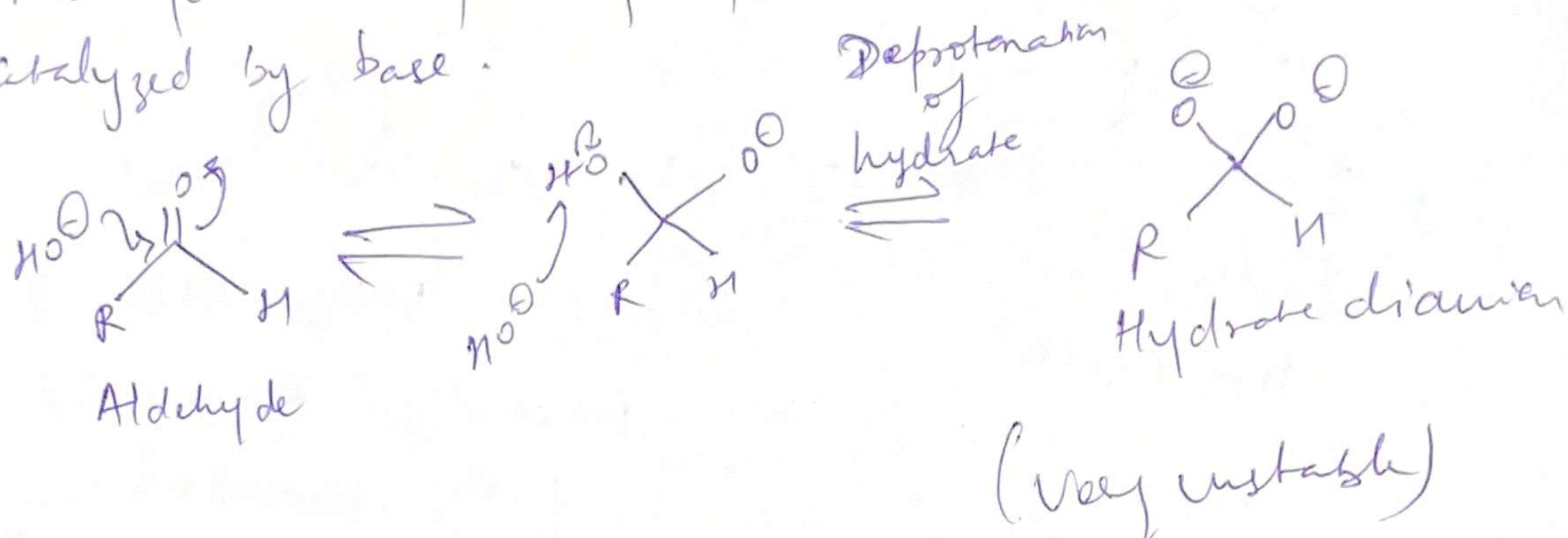
Very stable
Unenolized fraction is
poorly electrophilic

[Large amount of stable enol & a small
fraction of unenolized compound]



The Cannizzaro Reaction

Aldehydes are partly hydrated in water which is catalyzed by base.



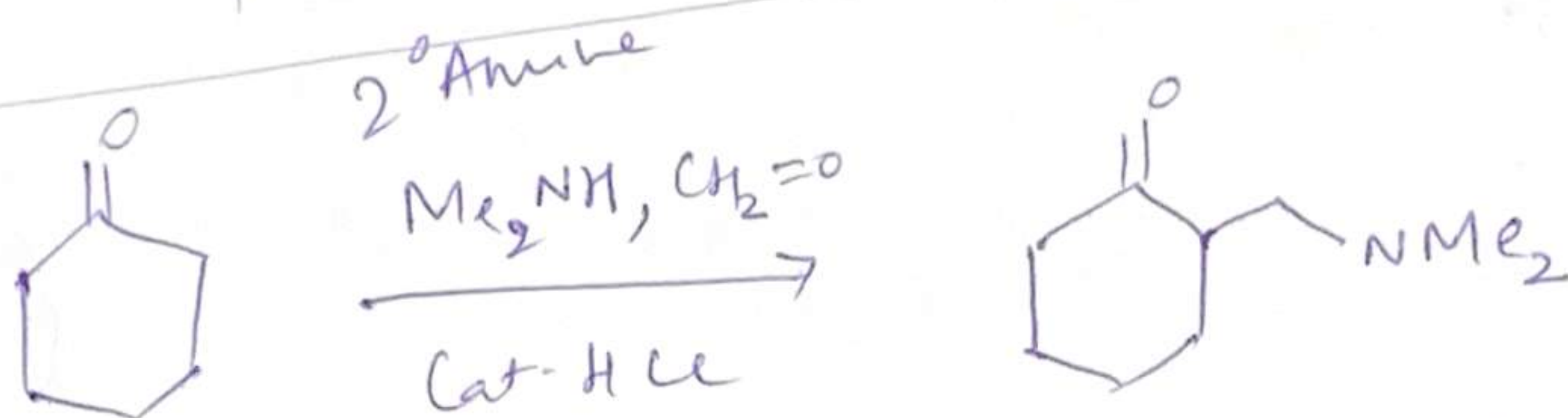
Disproportionation

Reaction works only with the unenolizable aldehydes

Mannich Reaction To make amino ketones ⑥ (drug molecules)

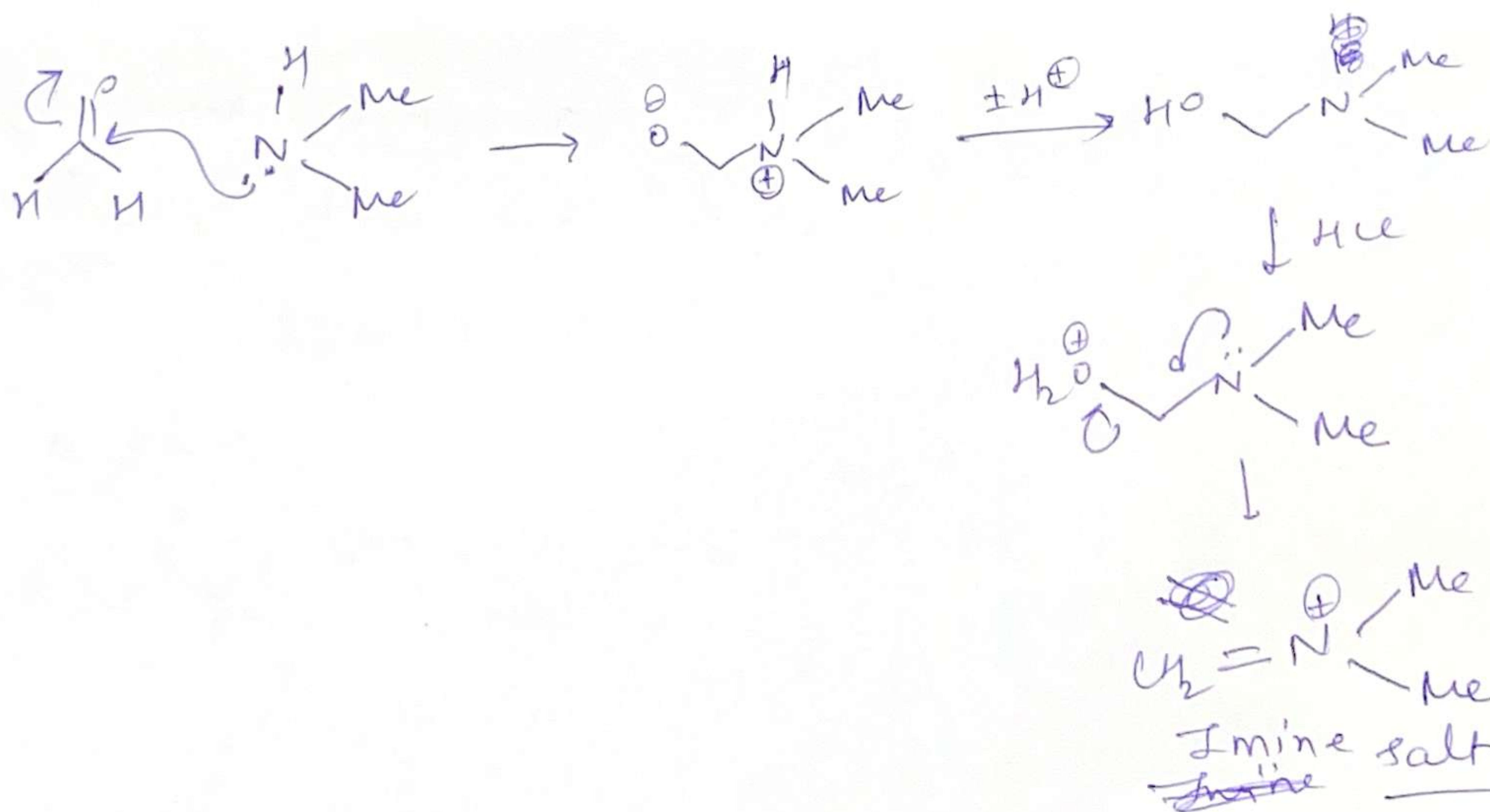
If we use formaldehyde as one of reactant in aldol, it can lead to Cannizzaro reacⁿ in the end.

In order to prevent that Mannich reacⁿ is done

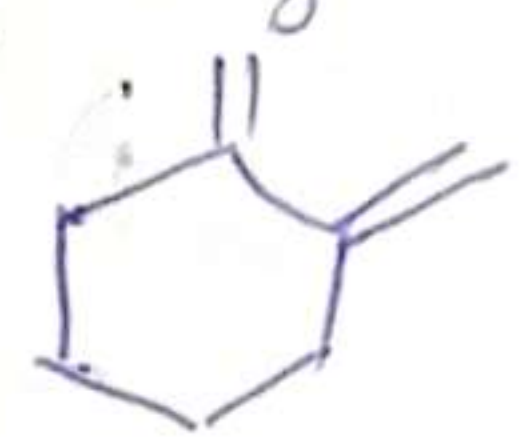
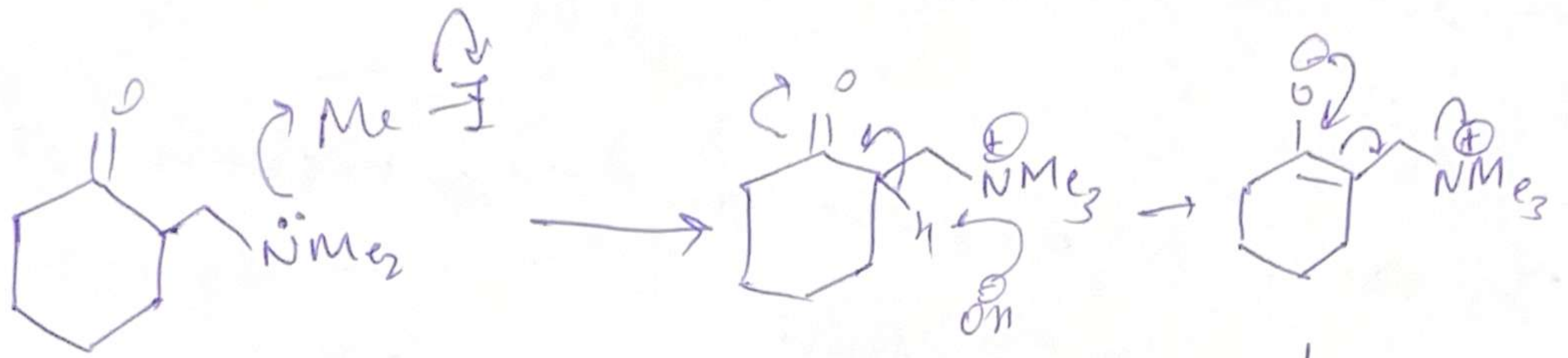
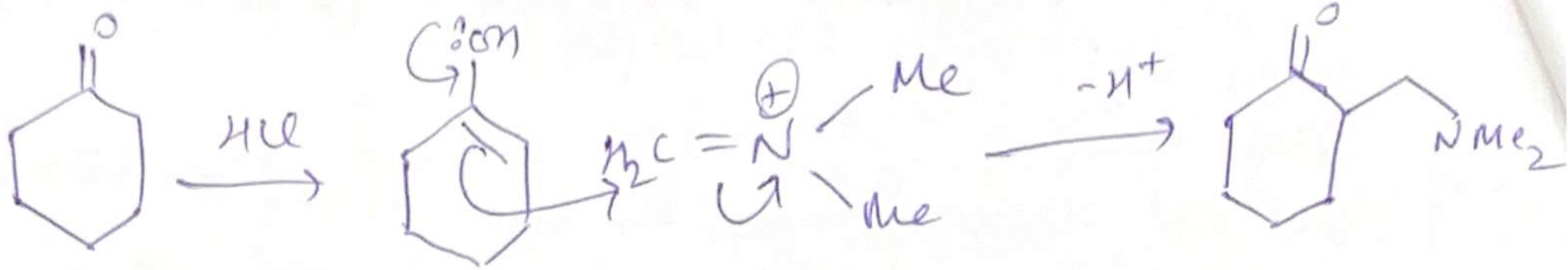


85%
yield

Enolizable
aldehyde or ketone



Corresponding iodide is sold as
Eschenmayer's salt



Exo-methylene
 compounds
 Very reactive

Cannot be made by
 aldol reacⁿ with formalde-
 hyde

Reformatsky Reaction

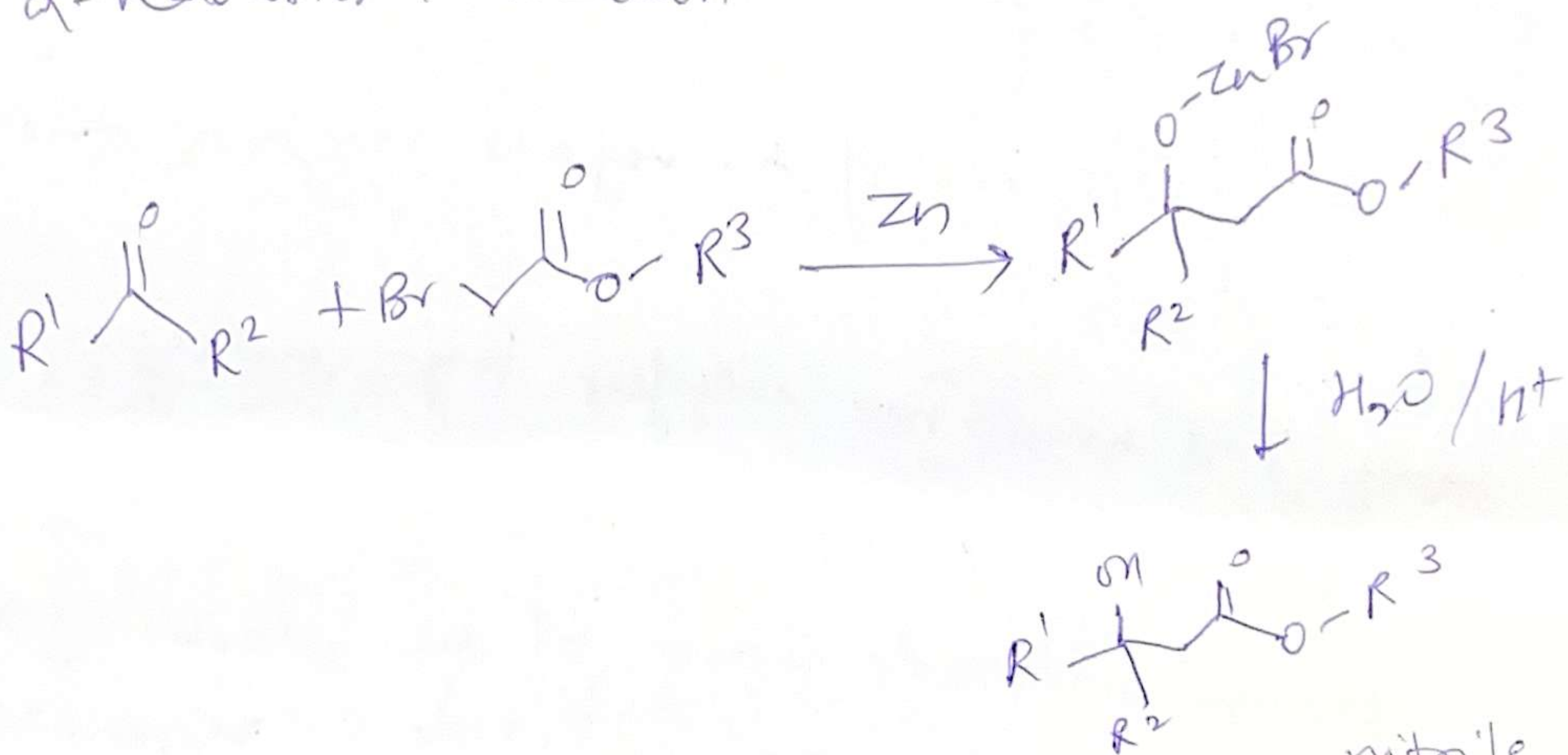
(+)

Aldehyde or ketones with α -halo esters using metallic Zinc to form β -hydroxy esters

(forms Zinc enolate)

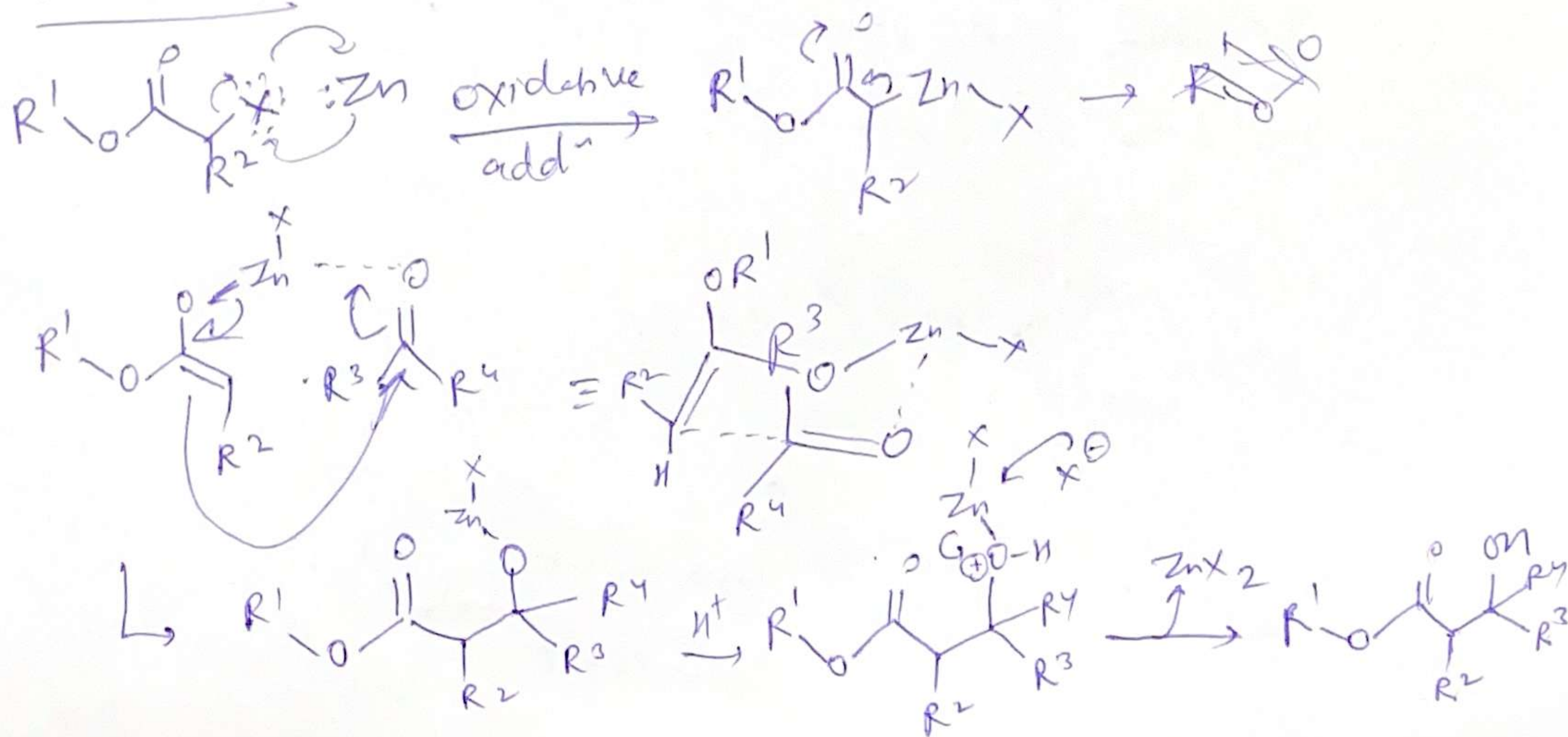
The organozinc reagent is called reformatsky ~~reagent~~ ^{reagent/} enolate

α -halo ester + Zn dust



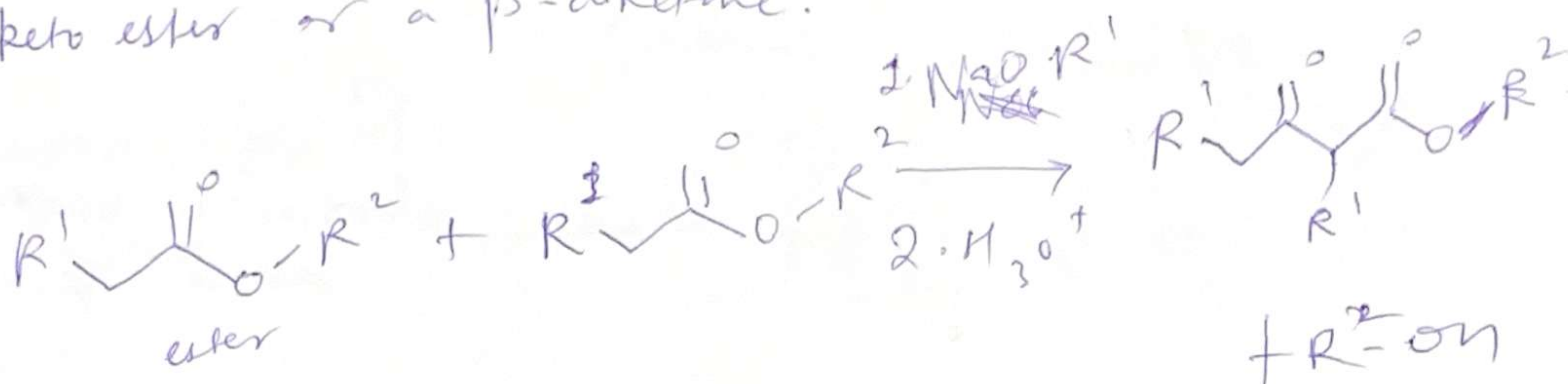
Substrate scope $\rightarrow$ Acid chloride, imide, ~~nitrate~~ ^{nitrile}, ketones

Mechanism



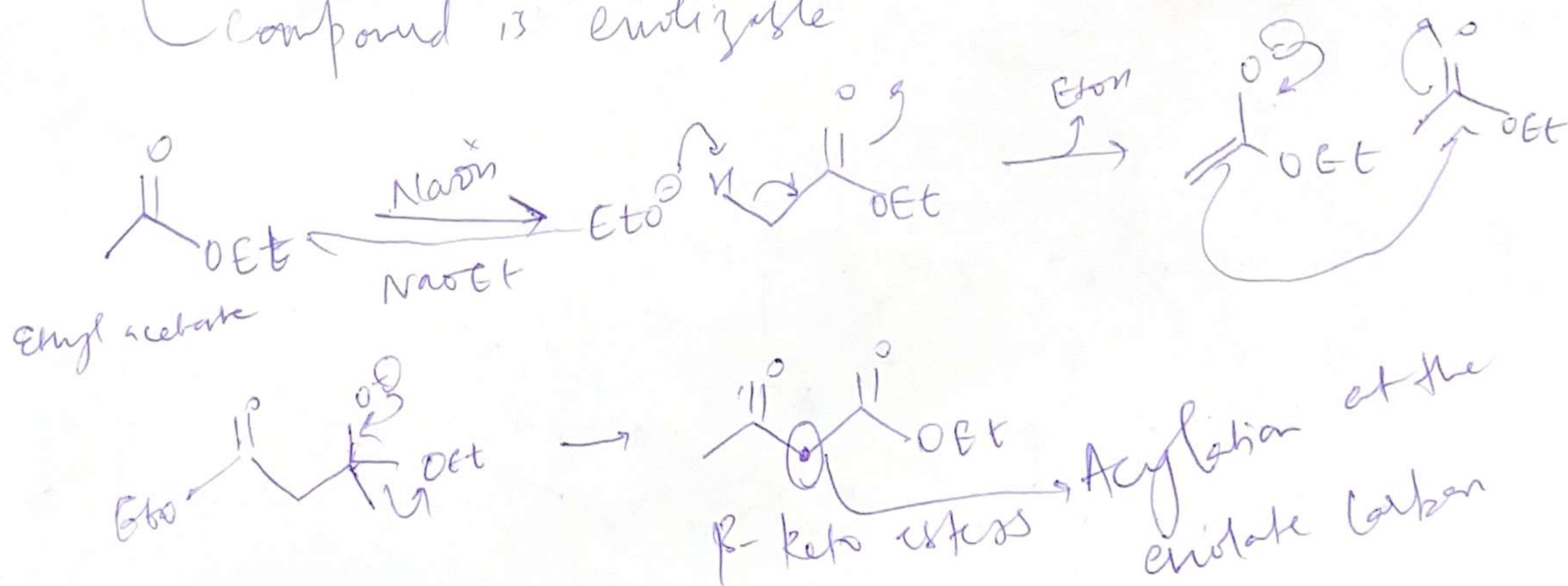
Claisen Condensation

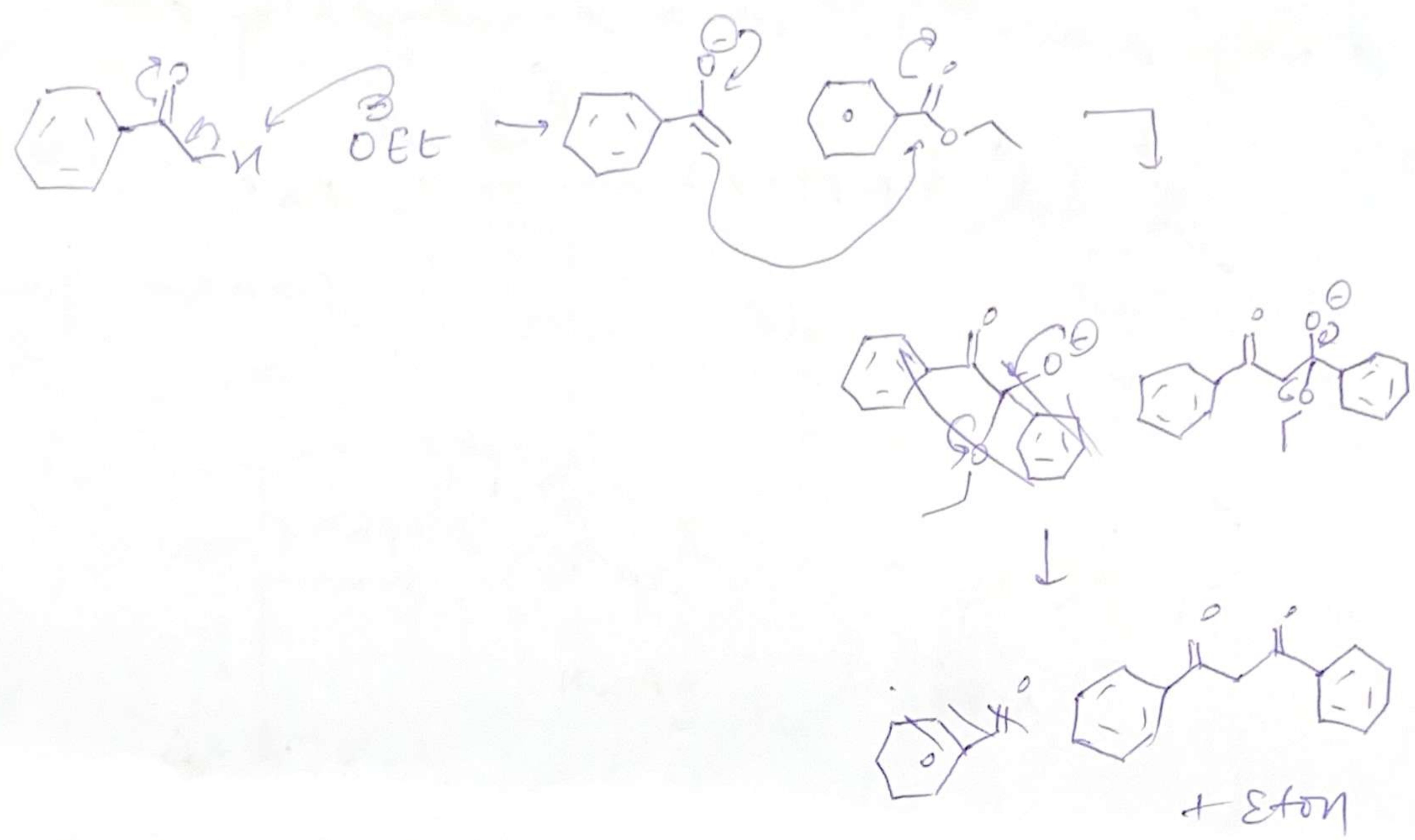
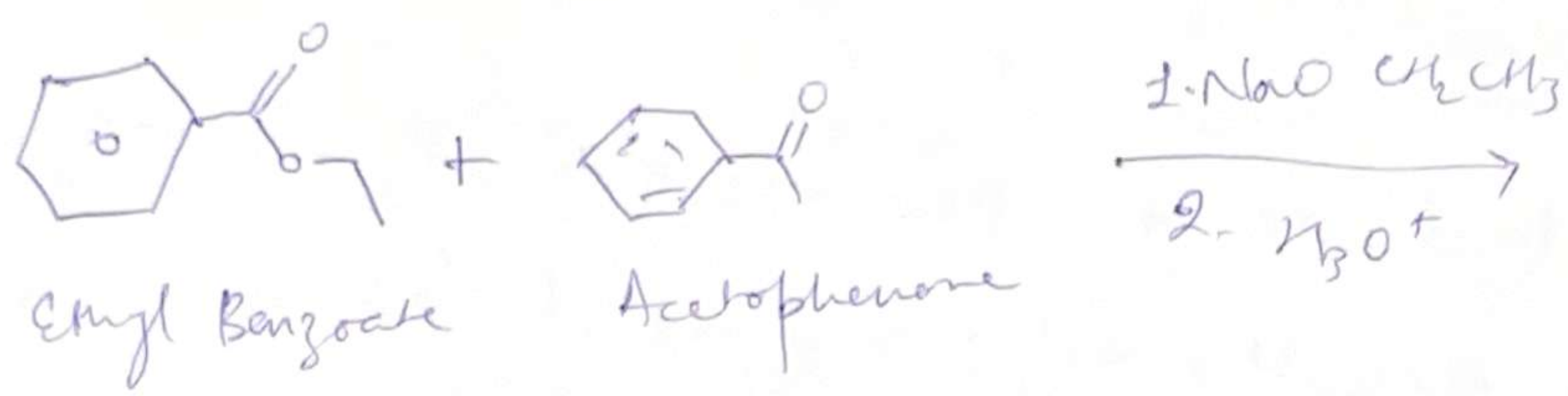
Between two esters or one ester & another carbonyl compound in presence of a strong base to produce a β -keto ester or a β -diketone.



- At least one of the reagents ~~should~~ ^{must} be enolizable.
- Base must not interfere by undergoing Nu substitution or addⁿ. Therefore, the conjugate sodium alkoxide base of the alcohol formed is used, since the alkoxide is regenerated.

[In mixed/cross claisen condensation non-nucleophilic base such as LDA, ~~some alkyne~~ compound is enolizable]





Dieckmann Condensation

Base-Catalyzed intramolecular reaction of a diester that forms a cyclic β -keto ester.

Works best for forming stable 5 & 6-membered rings.

